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REVIEW

Application of nanoindentation technology in rocks: a review

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Abstract Nanoindentation has become an increasingly popular method to determine the mechanical properties of both homogeneous and heterogeneous materials. Rocks are inherently heterogeneous and understanding their mechanical properties is of vital importance for relevant engineering applications. Due to its high precision and resolution in both force and displacement, nanoindentation can be utilized to extract the localized mechanical properties of individual grains. This review paper presents an overview of applications of nanoindentation techniques in various rocks, such as shale, coal, limestone, marble, sandstone and claystone. Apart from the conventional mechanical parameters, i.e., Young's modulus and hardness, other parameters, such as fracture toughness, time-dependent creep and tensile strength can also be obtained using nanoindentation methods. Basic equations and parameters employed to obtain the above mechanical parameters are clearly explained. In addition, merits and demerits of previous nanoindentation studies are summarised and roadmap for future trends of nanoindentation in geomaterials are suggested.

Keywords Nanoindentation · Geomaterials · Fracture toughness · Time-dependent creep

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1 Introduction

Understanding the mechanical properties of geomaterials (including rocks and soils) is of great significance to relevant engineering applications, such as reservoir stimulation (e.g. hydraulic fracturing, waterless-fracturing, in situ leaching), drilling, wellbore stability and rock physics modelling. Geomaterials are generally inherently heterogeneous with complex microstructures and mineralogical compositions at the micro-scale (Table 1). Macroscopic deformation as well as damage characteristics of geomaterials are dependent on the mechanical properties of their constituents (Li et al. 2019a). Although laboratory experiments can be carried out at meso- or macro-scale to obtain the mechanical parameters of geomaterials, such as Young's modulus (E) and hardness (H), these experiments surfer from the following four main disadvantages when compared with nanoindentation tests.

Table 1 Minerals constituents for different types of rock

Rock types	Constituents
Granite	Quartz, feldspar, mica
Shale	Quartz, feldspar, pyrite, illite, calcite, kaolinite, kerogen, dolomite
Coal	Quartz, clay, organic matter, vitrinite
Sandstone	Quartz, feldspar, clay
Claystone	Illite, kaolinite, quartz, calcite, dolomite, kerogen
Mudrock	Kerogen, clay,non-clay minerals

First, mechanical behaviour of geomaterials varies with mineral composition, particle type and mineral size (Yu et al. 2018); however, meso- or macroscopic laboratory tests do not provide the relationship among microstructural heterogeneity, the mechanical properties of individual phases and the deformation response (Manjunath and Jha 2019). To be more specific, meso- or macro-scale laboratory tests are not able to predict the mechanical parameters and failure behaviour of geomaterials samples with different compositions, sizes and anisotropies. By contrast, nanoindentation can be utilized to measure the mechanical properties of individual grains and microstructural units of material due to its high resolution in both displacement and load (Alcalá et al. 2009).

Second, conventional geomechanical tests need large core samples (Li et al. 2019b, c) while nanoindentation requires only a small volume of rock samples (rock chips) from drill cuttings, sidewall cores and fragments (Shukla et al. 2013), which is economical and less time-consuming.

Third, conventional geomechanical tests take ages in order to obtain the time-dependent creep behaviour and specimens are normally broken after the tests. In contrast, nanoindentation testing is fast and non-destructive and it has high measurement accuracy with loads in the 10^{-3} N range and the resolution is in the range of several 10^{-9} N. The penetration depth is on the order of microns with a resolution of less than 10^{-9} m (Fischer-Cripps 2011).

Forth, plasticity can be achieved for hard geomaterials easily at room temperature and rheology can be quantified in low-temperature deformation regimes because brittle deformation is suppressed due to the self-confined nature of the indentation method when compared to the conventional high-pressure and high-temperature rock deformation experiments (Kranjc et al. 2016).

As stated above, due to their high displacement and load resolution, real-time monitoring and high data acquisition rate (Huang et al. 2011), nanoindentation [aka instrumented indentation or depth-sensing indentation (DSI)] tests have been widely carried out to obtain the mechanical properties of materials at very small scales (i.e., micron and sub-micron scales). E, H and fracture toughness (K_{IC}) can be obtained from nanoindentation. These attributes are beneficial for the development of predictive rock physics models (Li et al. 2019b), the quantification of rock resistance to elastic as well as plastic deformation and for understanding the fracture initiation and propagation process (Gupta et al. 2018a; Bhushan 2017).

In addition, parameters like H/E and H^3/E^2 (Peng et al. 2012), time-dependent creep behaviour, elastic and plastic work, stress–strain curve (Pathak and Kalidindi 2015; Kalidindi and Pathak 2008; Pathak et al. 2009; Basu et al. 2006), brittleness (Fan et al. 2019), yield stress (Thom et al. 2017; Kang 2013), residual stress (Xu and Li 2008) and work-hardening exponent (Cheng and Cheng 2004) can also be obtained using nanoindentation methods (Lucca et al. 2010) (Fig. 1).

Normally, nanoindentation tests have three stages: loading, holding and unloading. The loading curve (OA) contains two different regions, the early elastic loading region (OA_1) and the late plastic loading region (A_1A), which indicates elastic and plastic deformation, respectively (Fig. 2). During the loading stage, the first pop-in event, i.e., incipient plasticity can be observed using a spherical tip (Ma et al. 2019a). The main deformation mechanisms related to the "pop-ins" are mainly due to the occurrence of crack formation, propagation or occurrence of chipping (Mikowski et al. 2008). The peak load P_m is reached at the end of loading stage. The holding stage (AB) is normally utilized to uncover the time-dependent deformation mechanisms of indented materials



Fig. 1 Functions of nanoindentation tests

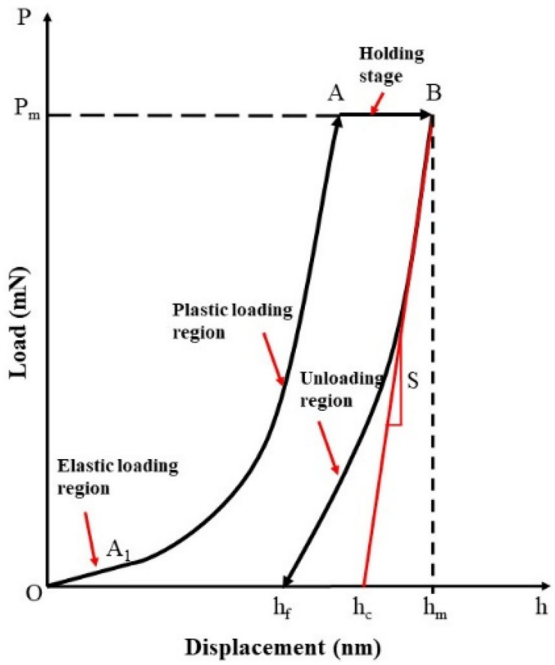


Fig. 2 Schematic of P–h curve during nanoindentation tests. *Note* h_f is the final depth of the contact impression after unloading and h_m is the maximum penetration depth of the indenter

(Goodall and Clyne 2006). The unloading stage (B-h_f) is the elastic recovery process and E and H values can be calculated using the Oliver and Pharr (O–P) method (Oliver and Pharr 1992, 2004). However, one dominant limitation for the O–P method is the accurate determination of the contact area because the contact area is normally measured indirectly (Eqs. 1–3).

Generally, pile-up phenomenon occurs for materials with low work-hardening rate, while sink-in occurs for materials with high work-hardening rate (Wang et al. 2012). To be more specific, pile-up occurs when E/σ_y is extremely low or n is close to zero while sink-in phenomenon becomes dominant when $n > 0.3$ (Jung-Min et al. 2012) (where σ_y is the yield strength and n is work hardening exponent). If pile-up happens, the contact area might be greatly underestimated (Wang et al. 2012); consequently, significant errors may be introduced on the measured E and H values (Tang et al. 2015; Pharr 1998). Alternatively, Huen et al. (2020) proposed the stiffness-based approach to derive the E and H values and this method is superior because it considered the sink-in and pile-up conditions while O–P method normally functions well only to material that has a sink-in contact profile. Similarly, Fang (2008) developed an electrical technique to measure in situ contact area to improve mechanical characterization at small scales because this method can determine the contact area directly while O–P method deduce the contact area from other parameters with some assumptions.

$$\mathrm{H} = \frac{P_m}{A} \tag{1}$$

where H is the hardness; P_m is the maximum force and A is the contact area.

$$\frac{1-v^2}{E} = \frac{1}{E_r} - \frac{1-v_i^2}{E_i} \tag{2}$$

where E, E_r and E_i are the Young's modulus of samples, reduced Young's modulus and indenter's Young's modulus, respectively; υ and υ_i are the Poisson's ratios for samples and the indenter, respectively.

$$S = \left(\frac{dP}{dh}\right)_{h=hmax} = \frac{2}{\sqrt{\pi}} E_r \sqrt{A} \tag{3}$$

where S is the stiffness and is equal to the slope of the initial unloading curve.

The nanoindentation technique was initially developed to investigate the mechanical properties of homogeneous materials (Borodich et al. 2015; Liu et al. 2018a). Although nanoindentation tests have

been carried out to investigate the mechanical properties of single crystals and polycrystalline aggregates, it has been underutilized for the geomaterials (Thom 2019). A single indent or a few indents cannot provide sufficient correct information about non-homogeneous materials; however, geomaterials are normally heterogeneous as they usually contain a plethora of phases and microstructures.

In order to investigate the mechanical properties of heterogeneous materials (e.g., geomaterials), one method is to combine a nano-indenter with optical or scanning electron microscopy (SEM), which can be utilized to locate the individual phase of interest. In order to obtain the mechanical properties of each mineral phase, the indenter size should be sufficiently smaller than the characterized mineral phase size (Abousleiman et al. 2010; Mason et al. 2014). Another method is to perform a number of equidistantly-spaced indentations and undertake a statistical evaluation (Haušild et al. 2016, 2017). This method is known as grid indentation (or statistical nanoindentation) and is a 2-dimension-mapping tool for examining the properties of constituent phases independently of each other (Du and Radonjic 2018).

Grid nanoindentation was initially invented by Constantinides et al. (2003, 2006) to study the mechanical properties of heterogeneous materials through developing a map of the chemo-mechanical constituents of materials (Zhu et al. 2009; Abedi et al. 2016a; Akono and Kabir 2016; Hou et al. 2016). In 2006, Ulm and Abousleiman (2006) first utilized the grid nanoindentation method to examine highly heterogeneous shale samples, followed by other researchers who adopted grid indentation to extract the mechanical properties of various geomaterials such as rocks, soils and sand (Li et al. 2018, 2019a, c; Abedi et al. 2016a; Bobko 2008; Mba and Prasad 2010; Deirieh et al. 2012; Kumar et al. 2012a; Bennett et al. 2015; Liu et al. 2016, 2017a, b, 2018a, b; Liu and Ostadhassan 2017; Veytskin et al. 2017; Yang et al. 2017; Vialle and Lebedev 2015; Zhang et al. 2018a; Zargari et al. 2011; Cała et al. 2017; Tankiewicz 2018; Akono et al. 2019; Kabir and Akono 2018).Although grid nanoindentation can provide information on all phases' properties and their volume ratios, the analytical result should be compared with the findings from the SEM observation and/or X-ray diffraction analysis due to the limitations and accuracy of the grid nanoindentation and statistic deconvolution method (Hrbek and Koudelková 2017).

2 Nanoindentation testing of various geomaterials

Nanoindentation tests have been performed to investigate the mechanical properties of various geomaterials, such as shale (Fig. 3), clay (Hou et al. 2016; Pant 2013; Berthonneau et al. 2017; Zhang et al. 2013; Ma

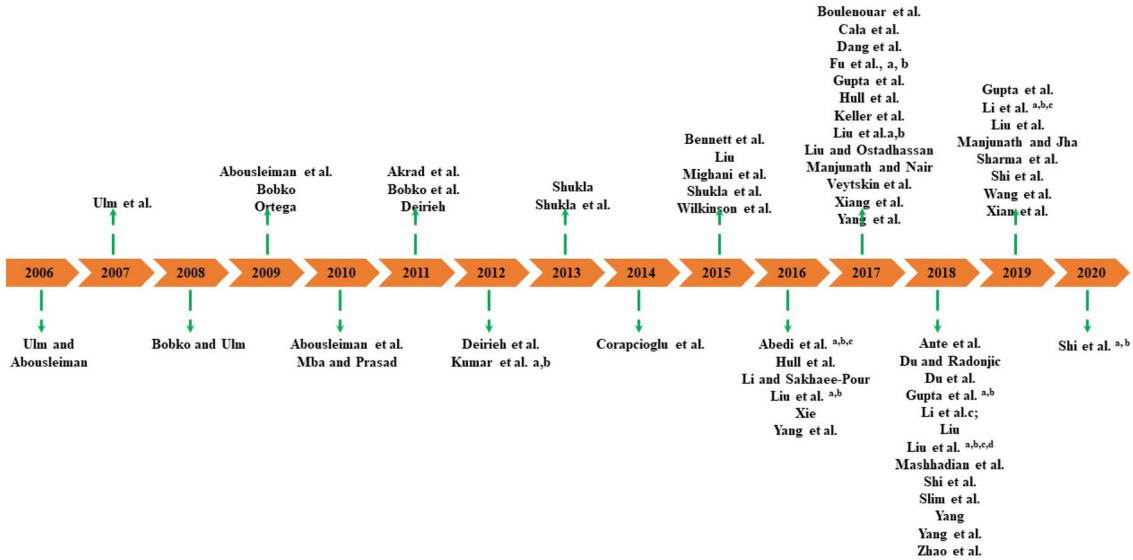


Fig. 3 Timeline for application of nanoindentation to shale samples

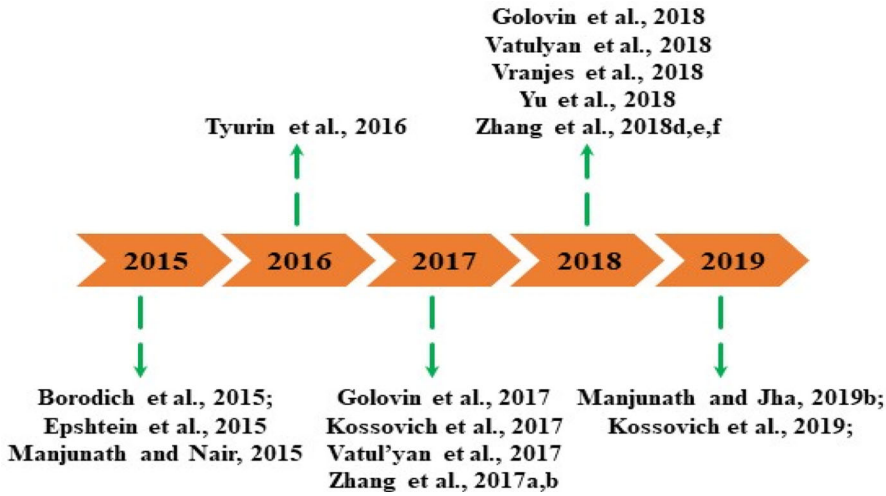


Fig. 4 Timeline for application of nanoindentation to coal samples

et al. 2019b; Pant et al. 2013; Yin and Zhang 2011a; Basu et al. 2009), coal (Fig. 4), sandstone (Zhu et al. 2009; Akono et al. 2019; Kabir and Akono 2018; Bobko 2006; Ante et al. 2018; Bastos de Paula 2011; Tyurin et al. 2016; Saenger et al. 2016; Shukla et al. 2015; Viktorov et al. 2014; Golovin et al. 2017, 2018), limestone (Vialle and Lebedev 2015; Shukla et al. 2015; Ban et al. 2014; Zhang et al. 2016, 2018b; Lebedev et al. 2014a, b, 2017; Vialle et al. 2018; Kalo 2017), marble (Tyurin et al. 2016; Golovin et al. 2017, 2018; Bandini et al. 2012, 2014; Konstantinidis et al. 2016; Brooks et al. 2010; Sly et al. 2019), claystone (Zhang et al. 2018a; Magnenet et al. 2011; Auvray et al. 2017; Bartier and Auvray 2017), soil and sand (Dutta and Penumadu 2007; Kadali et al. 2013; Nadukuru 2013), 3-D printed rocks (Kong et al. 2018, 2019), olivine (Kranjc et al. 2016; Kumamoto et al. 2017; Kranjc 2017), granite (Tyurin et al. 2016; Viktorov et al. 2014; Golovin et al. 2017, 2018; Zhang et al. 2018c), varved clay (Tankiewicz 2018), mudrocks (Valdes and Heidari 2017), quartz (Goldsby et al. 2004; Broz et al. 2006; Thom et al. 2018; Sahoo et al. 2017), halite (Thom and Goldsby 2019), natural quartzite and granodiorite (Zhu et al. 2007), etc. (Mikowski et al. 2008; Ashraf et al. 2015; Hansen et al. 2020; Whitney et al. 2007; Deng 2017; Qiu 2019; Beirau et al. 2016). According to Figs. 3 and 4, the application of nanoindentation testing on geomaterials is relatively new and most studies were carried out in the last decade. In addition, among various types of geomaterials, only shale rock is relatively well studied and the main attention has been focused on fluid-shale interactions (Liu et al. 2018a; Akono et al. 2019; Kabir and Akono 2018; Lebedev et al. 2014a, b; Yang et al. 2016; Corapcioglu et al. 2014; Du et al. 2018; Yang et al. 2018; Mighani et al. 2018a; Wang et al. 2019a; Gupta et al. 2017, 2018b, 2019; Akrad et al. 2011) and creep characterization (Gupta et al. 2018a; Mighani et al. 2015; Liu et al. 2018c, d, 2019a; Slim et al. 2018; Sharma et al. 2019) apart from the common mechanical parameters (say E and H).

2.1 Shale

Shale is composed of various constituent phases, such as carbonate and quartz and the brittleness of which is greater than that of clay minerals (Goral et al. 2018). Many researchers have applied nanoindentation tests to measure the mechanical properties of shale samples (Fig. 3). For instance, Ahmadov et al. (2009) found the E and H values of kerogen in Bazhenov Formation oil shale to be 10–15 GPa and 570 MPa, respectively, which are consistent with the findings of Zeszotarski et al. (2004) for Woodford shale. Shukla et al. (2013) also measured the E of kerogen (ranging from 5 to 9 GPa).

Since inorganic crystals such as pyrite, dolomite, calcite and quartz are stiffer and stronger than kerogen (Eliyahu et al. 2015; Wilkinson et al. 2015; Boulenouar et al. 2017), some researchers have investigated the impact of the percentage of total organic carbon (TOC) and clay minerals on the mechanical parameters of shales. For instance, Li et al. (2018) found that E decreases with the increase of TOC and clay content,

which is consistent with (Li et al. 2019b; Shukla et al. 2013; Kumar et al. 2012a, b, 2015; Xie 2016; Liu 2015, 2018; Zargari 2015; Wang and Zhang 2018; Shukla 2013; Shi et al. 2018). TOC has a dominant role in the mechanical properties of shale rocks and TOC reduces with thermal maturation; the impact of thermal maturation on the mechanical properties of shale rocks have been widely studied (Shukla et al. 2013; Zeszotarski et al. 2004; Ahmadov et al. 2009; Kumar et al. 2012b; Bobko and Ulm 2008; Zargari et al. 2013; Alstadt et al. 2016; Alsinan 2017; Slim 2016) using nanoindentation. Xie (2016) found that the E of kerogen increased with increasing thermal maturation, which is in agreement with (Alsinan 2017; Slim 2016). However, Zargari (2015), Zargari et al. (2016) reached the opposite conclusion that the kerogen underwent modulus reduction with increasing thermal maturation due to the existence of gas- or bitumen-filled pores.

Nanoindentation tests have also been carried out to investigate the anisotropy properties of shale samples. For instance, Liu et al. (2017b) found that the mechanical parameters (i.e. E and H) parallel to the bedding line differ from the values perpendicular to the bedding line, indicating the anisotropy properties, and these findings are in consonance with (Shukla et al. 2013; Shukla 2013; Shi et al. 2018). The anisotropic behaviour of shale rocks is affected by the transversely isotropic symmetry of clay (Ortega et al. 2009). Other researchers have also investigated the impacts of the amount of TOC and clay on the anisotropy of shale rocks. For example, Li et al. (2019b) found that mechanical anisotropy increases with increasing TOC content, which coincides with the findings of Kumar et al. (2012a).

In addition, some researchers have compared the anisotropy differences of E, H and K_{IC}; for instance, Xiang et al. (2017) found that the E of shale samples had a small degree of anisotropy while the K_{IC} exhibited a much higher degree of anisotropy. Cała et al. (2017) found that the E_r was anisotropic while there existed no anisotropy characteristics for the H. However, Boulenouar et al. (2017) found that there was no difference between the results of nanoindentation tests on surfaces parallel or normal to the bedding and they surmised that this was because the anisotropy had been obscured by the heterogeneity, which is in agreement with Fu et al. (2017), Fu (2018). Zeszotarski et al. (2004) also concluded that there was no detectable anisotropy in kerogen mechanical properties, and they argued that anisotropy was closely correlated to overburden pressure and surface annealing.

Some researchers have also investigated the relationship between the measured mechanical parameters, namely E, H and K_{IC}. For instance, Liu et al. (2016, 2017a), Liu and Ostadhassan (2017) and Liu (2018) observed that K_{IC} increased with E, which is in agreement with (Ostadhassan et al. 2018; Xian et al. 2019). Ante et al. (2018) found that H was proportional to the E with an almost linear relationship for both shale and sandstone, which is in line with Fu et al. (2017), Fu (2018).

Apart from the determination of E, H and K_{IC}, the creep behavior of shale rocks has also been studied. Liu et al. (2018c) found that the displacement increased as the creep time increased; however, in a later study, they (Liu et al. 2019a) found that abnormal behavior occurred during the nanoindentation holding stage of shale rocks. More specifically, the displacement first increased with the creep time and after a critical creep time, the displacement decreased. They concluded that creep behavior dominated the process at the beginning of the holding period while elastic recovery played a more significant role as the holding time proceeded. Apart from the impact of holding time (i.e., creep time or dwell time), the impact of constituent phase contents on creep behavior has also been investigated. Gupta et al. (2018a) found that the time-dependent creep increased as a function of clay content, porosity and TOC, and decreased as a function of quartz and carbonate content.

In addition to pure nanoindentation tests, other researchers have combined nanoindentation with other experimental techniques to characterize the material phases. For instance, energy dispersive spectroscopy (EDS) has been utilized to measure all the chemical elements within the probed area and spatial elemental maps can be adopted to constrain the mineralogy at indentation sites when combined with nanoindentation (Abedi et al. 2016a, b; Veytskin et al. 2017; Boulenouar et al. 2017; Mashhadian et al. 2018). Apart from EDS, other methods, such as X-ray microtomography (CT) (Cała et al. 2017), back-scattered electron diffraction (BSED) (Shukla et al. 2013; Kumar et al. 2012a, b) and confocal laser-scanning microscopy (CLSM) (Ahmadov et al. 2009) have also been adopted to investigate the individual material phases

at small volumes of indented materials. Moreover, some microscopy, such as scanning electron microscope (SEM), transmission electron microscope (TEM) and scanning probe microscope (SPM) have been adopted to investigate the deformation mechanisms of materials (Cui et al. 2018). The sink-in/pile-up phenomenon can be observed using optical microscopy, SEM, interferometry, atomic force microscope (AFM) and confocal microscopy, depending on the size of the residual indentation imprint.

Currently, both hydraulic and waterless fracturing are adopted in unconventional reservoirs (Yang et al. 2019, 2020) and fluid-rock interactions are of great importance to hydraulic-fracture conductivity and consequently, reservoir productivity. However, applications of nanoindentation testing in fluid–rock interactions have been relatively infrequent and much more attention should be paid to this area. The wettability of shale formations is of vital importance to reservoir production. Lu et al. (2019) found that the E of clay matrix decreased from 29.2 to 16.5 GPa after it reaction with water for 30 days. When shale rock is water-wet, the reduction in $E$and H is larger, while when the shale rock is oil-wet, the reduction in E and H is smaller (Gupta et al. 2017, 2018b2019). Similarly, Yang et al. (2016, 2018) investigated the impact of surface treatments on the mechanical parameters of shale samples. It was found that shale samples with hydrophobic surfaces exhibited a much lower degree of softening than untreated samples.

Corapcioglu et al. (2014) found that the occurrence of KCl-based fluids resulted in a reduction in E of shale samples and they assumed that the main reason for this was the dissolution of calcite mineral, which is in agreement with Du et al. (2018). Similarly, Gupta et al. (2017, 2018b, 2019) investigated the impacts of brine and dodecane on E and Hin shales, and they found that brine resulted in a greater reduction in E than dodecane. In addition, the increase in clay stabilizer concentration led to lower weakening. Wang et al. (2019a) investigated the impact of supercritical CO_2 ($scCO_2$) flooding on the mechanical properties of Bakken shale and found that after $scCO_2$ injection, the E of samples decreased. Shi et al. (2020a) found the similar phenomenon that the shale samples treated by $scCO_2$ demonstrated lower H, E and K_{IC}, which is in agreement with Tian et al. (2020), Zhang et al. (2019), who also found that both the E and H for shale sample decreased significantly after it reaction with carbonic acid and the rock sample becomes more ductile after acid treatment.

Apart from the impacts of brine and $scCO_2$, the influences of other factors, such as loading rate (Shi et al. 2019), penetration depth (Bennett et al. 2015; Viktorov et al. 2014), and temperature (Corapcioglu et al. 2014; Akrad et al. 2011) on mechanical properties have also been studied. Shi et al. (2019) found that the H, E, yield stress and indentation shear stress of shale increased with the loading rate. Viktorov et al. (2014) found that H reduced with penetration depthand they also found that Edecreased for all rock-forming minerals until the depth of 500–600 nm and remained the same with further increase of penetration depth. This conclusion is in agreement with Li et al. (2019a), Bennett et al. (2015) and Cała et al. (2017). Corapcioglu et al. (2014) studied the influence of temperature on the E of shale samples. They found that samples heated to a higher temperature had a greater reduction in E than samples exposed to KCl solutions at room temperature because the higher temperature accelerated the dissolution rate; this is in agreement with the findings of Akrad et al. (2011). We conclude from these studies that at shallow depths, the measured mechanical properties correspond to the deformation response of the constituent materials; while at deep depths, the measured mechanical properties correspond to the deformation response of the composite matrix because of the disturbance of the surrounding constituents and the supporting matrix. These findings are also validated by studies Constantinides et al. (2006), Shukla et al. (2015) and Zargari et al. (2013) of the impact of grain size on nanoindentation results. It was found that when the grain size is large (coarse-grained rocks), the measured E is the grain E because of the relatively small ratio of indentation area to grain size. For finer-grained rocks, such as shales and pyrophyllite, the indention area is larger than the average grain size; hence, the measured E are representative of the rock Young's modulus. In other words, nano-scale indentation occurs when the indentation depth is of the order of a few hundreds of nanometers, while micro-scale indentation occurs when the indentation depth is of the order of a few thousands of nanometers (Mason et al. 2014; Yang et al. 2017; Cała et al. 2017), this is in agreement with ISO 14577-1 (Lucca et al. 2010), as shown in Table 2.

Table 2 Application ranges of nanoindentation testing. (adapted from Lucca et al. 2010)

Application range	Corresponding conditions
Nano range	$h_{max} \leq 200$ nm
Micro range	$h_{max} > 200$ nm and $P_{max} < 2$ N
Macro range	2 N $\leq P_{max} \leq 30$ kN

h_{max} is the maximum indentation depth and P_{max} is the maximum test force

This conclusion supports the assumption of grid indentation. However, a different conclusion was reached by Han et al. (2018), who found that the E and H of shale increased rapidly with the increase of indentation displacement and decreased rapidly after reaching a certain value. With the sustained increase of penetration depth, the E and H of shale became stable, and finally, the E and H of shale decreased by degrees with the increase in indentation displacement.

Nanoindentation tests are normally performed to investigate E, H, K_{IC} and the time-dependent creep behaviour. Recently, nanoindentation techniques have been utilized to study the tensile strength of rocks by loading and failing micro-cantilever geometries (Abousleiman et al. 2016; Hull et al. 2017) because of the advancement of sample preparation. Furthermore, Keller et al. (2017) determined the micromechanical properties of clay matrix by compressing the micropillars with an in situ SEM indenter and found that the porous clay matrix behaved more rigidly when a load was applied parallel to bedding than when load was applied perpendicular to bedding.

2.2 Coals

Similar to shale rocks, coals are also composed of various phases (Table 1) and the internal structural features of coals span multiple length scales (Kossovich et al. 2019). To put it another way, they are heterogeneous and hierarchical composite materials. Kossovich et al. (2017) measured the E and Hof three different coals and found that the E of coal matrix decreased slowly from lignite to medium-rank bituminous coal while the E of coal inclusions decreased significantly from lignite to bituminous coal. For H, it was found that for the coal matrix, the hardness increased first from lignite to low-rank bituminous coal then decreased from low-rank bituminous coal to medium-rank bituminous coal while the H of inclusions decreased from lignite to bituminous coal with the increase of coal rank. For nanoindentation, thin transparent films of coal were prepared in order to avoid the impacts of pores and cracks and transmitted light microscopy was adopted to position the indent probe and identify the micro-components (Kossovich et al. 2017). This method can evaluate the E and H of coal samples effectively although the thickness of glue may induce some changes to the effective properties of materials (Epshtein et al. 2015). Apart from thin films of coals, other methods have also been adopted to combine the nano-scale geomechanical properties with microstructural morphology and composition, such as EDS and XRD (Yu et al. 2018) and micro-CT (Zhang et al. 2017).

Similarly, Zhang et al. (2018d) investigated the impact of coal rank on the mechanical properties of five types of coal (anthracite, bituminous, sub-bituminous, lignite, and peat), and found that the coal indentation moduli decreased from high- to low-rank coal. They compared the mechanical properties measured by nanoindentation tests with the results obtained from traditional acoustic methods (Zhang et al. 2018e, f). It was found that the minerals had higher moduli while the coal matrix had the lowest moduli and they also concluded that the nanoindentation test is superior because the bulk acoustic method cannot identify the weak spots. Apart from the influence of coal ranks, the effect of water content on mechanical properties has also been investigated. Zhang et al. (2018f) found that the measured indentation moduli decreased after water adsorption, which is different from previous acoustic tests.

2.3 Limestone

Limestone is a kind of carbonate sedimentary rock and the majority (> 90%) of composite materials are calcite and aragonite, which are polymorphisms of $CaCO_3$. Porous limestone is a potential host rock for CO_2 sequestration. Because limestone contains a wealth of $CaCO_3$, it is reactive and sensitive to acidic fluids. Therefore, Zhang et al. (2016) investigated the change of mechanical properties before and after exposure to $scCO_2$. They found that after $scCO_2$ injection, the E_r became higher for low indentation modulus areas. In contrast, for high indentation

modulus areas, the E_r after $scCO_2$ injection were lower than those before $scCO_2$ injection. Similarly, Lebedev et al. (2014a, b, 2017) compared the E_r change for dried, water-saturated, hydrocarbon-saturated and brine-flooded samples. It was found that the rock matrix softened significantly by acid exposure and water saturation, and the E_r for the area inside the grain was different from that close to a grain boundary (even for a homogeneous grain) because the grain is bonded to other grains and the bonding may decrease or increase the rigidity of the grain. Mighani et al. (2018b) found that both the E and H values decreased after the water saturation for carbonate. In addition, the creep deformation increased when compared with the dry indentations. However, for oil-saturated samples, both the E and H increased while the creep magnitude decreased.

Due to the heterogeneity of the limestone sample, Zhang et al. (2018b) found that the E_r ranged from 1 GPa to 40 GPa, which is in agreement with (Zhang et al. 2016; Lebedev et al. 2014a). De Paula (2011) measured the E of oolitic limestone and found a large variation in calcite bulk modulus from 56 to 144 GPa due to the influence of micro-porosity. This conclusion has been validated by other researchers (Kalo 2017)(Vialle and Lebedev 2015) who found that macro-calcite is hardest while micro-calcite is weakest because of the impact of porosity.

Marble is limestone that has been recrystallised and is suitable for the study of brittle fractures because it is a relatively pure, homogenous, and isotropic brittle material (Brooks et al. 2012). Brooks et al. (2010, 2012, 2013; Brooks 2010) compared the E_r of intact marble and the fracture process zone (FPZ) and found that the FPZ had a low reduced indentation hardness and E_r. In addition, they studied the impact of grain size on the FPZ and found that the FPZ increased with grain size.

2.4 Sandstone

Apart from coal and limestone, sandstone is another type of host formation for CO_2 sequestration (Akono et al. 2019; Kabir and Akono 2018). Different from limestone, sandstone is mainly composed of quartz while the limestone is dominated by calcite. During the injection of CO_2 and after the geochemical reaction for a long time, sandstone aquifer will be more stable than that of limestone (Raza et al. 2019). In addition, sandstone is normally adopted for buildings (Zhu et al. 2009; Koudelka et al. 2014). Sandstone is also an important reservoir rock and the hydraulic fracturing process normally occurs at the sandstone formation (Ante et al. 2018). Therefore, understanding the mechanical properties of sandstone is vitally important. For instance, the fracture initiation and propagation of sandstone is closely correlated with the Young's modulus, hardness and fracture toughness (Ante et al. 2018).

Akono et al. (2019), Kabir and Akono (2018) investigated fluid-rock interactions and nanoindentation tests were performed to observe the variation in chemical micro-constituents. It was found that porosity increased and K-feldspar decreased after interaction with fluids, indicating a dissolution reaction involving feldspar. Maruvanchery and Kim (2020) and Maruvanchery (2019) investigated the impact of temperature on the elastic properties of sandstone and they found that the E_r and H of the minerals and mineral-to-mineral contacts reduced greatly in furnace-heated samples when compared with the oven-dried samples.

2.5 Claystone

Understanding the mechanical properties of claystone is of great significance to the geological storage of nuclear wastes (Magnenet et al. 2011) and CO_2 sequestration where claystone acts as caprock. Auvray et al. (2013a, b, 2015) investigated the impact of water on the mechanical properties of argillite and found that the modulus decreased with saturation. Apart from the relative humidity levels, Arnold et al. (2015) also investigated the influence of temperature on the mechanical properties of claystone and found that collapse of hardness occurred above 90% relative humidity at 60 °C.

In addition, the impacts of the scaling effect and alkaline fluid on the deformation modulus of claystone were also studied by combining nanoindentation, micro-indentation and meso-compression testing (Auvray et al. 2017; Bartier and Auvray 2017). It was found that the variation of elastic modulus is due to changes in the structure and mineralogy within a sample. Similarly, some researchers have carried out upscale studies and Mori–Tanaka (MT), Self-Consistent Scheme (SCS), Differential Effective Medium (DEM) theory have been utilized for upscaling the

microscale mechanical properties to macroscale ones (Kong et al. 2019). For instance, Zhang et al. (2018a) adopted two homogenization methods to determine the macroscopic Young's modulus of claystone according to the Young's modulus of claystone measured by nanoindentation methods.

2.6 Clay mineral (phyllosilicate)

Clay minerals are abundant in soils, shales and geological high-level radioactive waste repositories (Ma et al. 2019c) and are characterized by anisotropic properties due to their distinct, nano-scale layered crystal structure (Pant et al. 2013). The particle size of clay minerals is relatively small and modulus mapping have higher resolution compare with the nanoindentation method; hence, modulus mapping is more suitable for the characterization of clay minerals (Zhao et al. 2018).

The measured E and H values can be affected by various factors, such as hold time, loading rate, maximum load (indentation depth) and cyclic loading times. Zhang et al. (2009) compared the behaviour of two clay minerals (muscovite and rectorite) and found that rectorite exhibited a more highly time-dependent response and had much lower E and H values than muscovite. In addition, an indentation size effect was observed for muscovite during nanoindentation experiments (Zhang et al. 2009, 2010). In other words, E and H values decreased with increased indentation depth.

2.7 Sand and soil

Sand is a type of granular material and its size ranges from millimetres to micrometres. Understanding the mechanical properties of sand is of vital importance to geotechnical engineering fields, such as landslides, tectonic plate sliding and rock-fill dams (Dutta and Penumadu 2007; Daphalapurkar et al. 2011). Nanoindentation tests have been performed to investigate the mechanical properties of sand particles, such as E, H (Dutta and Penumadu 2007; Nadukuru 2013; Daphalapurkar et al. 2011; Erdoğan et al. 2017; Xu et al. 2018; Luo et al. 2015; Hu et al. 2014; Wang et al. 2011, 2014; Penumadu et al. 2009; Komanduri et al. 2011), K_{IC} (Daphalapurkar et al. 2011; Xu et al. 2018; Wang et al. 2011, 2014; Komanduri et al. 2011) and creep behaviour (Nadukuru 2013). Wang et al. (2011) investigated the impact of grain size on the mechanical properties of sand grains and found that larger sand grains had larger E and H, which is consistent with Komanduri et al. (2011).

The mechanical properties of soil have also been investigated. Previous studies have mainly focused on the impact of temperature on the mechanical properties of soils (Kadali et al. 2013; Singh 2019; Park 2018). It has been found that an obvious change occurs in the H, residual modulus and resistance to indentation of soils subjected to high temperatures, and the impact of temperature is more prominent on fine-grained soils than on coarse-grained soils.

3 Fracture toughness calculation

Fracture toughness (K_{IC}) (aka critical stress intensity factor) is an essential mechanical parameter indicating resistance to crack propagation, and the accurate measurement of K_{IC} is of vital importance to hydraulic fracturing. Normally, large specimens are needed for conventional standardized methods to determine K_{IC} in order to satisfy the plane strain condition. Nanoindentation can also be adopted for the accurate measurement of fracture toughness and this method only requires a small sample size (Feng and Zhang 2015). In this section, we review four common nanoindentation methods for the study of fracture toughness of geomaterials.

3.1 Crack-length-based method

In this method, K_{IC} is calculated according to the length of the cracks and it is correlated with crack system types and the geometry of the pyramidal indenter. Normally, fracture toughness based on pyramidal indentation tests is expressed as:

$$K_{IC} = \frac{P_{max}}{c^{3/2}} \prod \left(\frac{E}{\mathrm{H}}, v, \psi, \frac{c}{a} \right) \tag{4}$$

where K_{IC} is the fracture toughness, P_{max} is the maximum indentation load, E is the elastic modulus, H is the hardness, v is the Poisson's ratio, ψ is the axis-to-face angle of the pyramidal indenter and c and a are the average crack length and the contact size (the distance from the centre to the indentation to the corner of the contact), respectively (as shown in Fig. 5) (Sebastiani et al. 2015a).

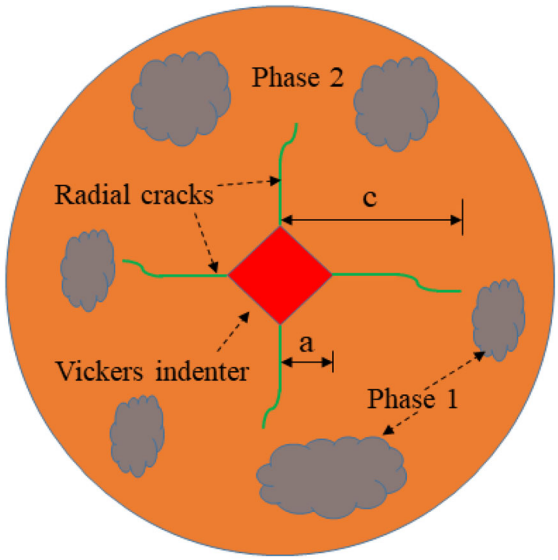


Fig. 5 Schematic illustration of radial cracking at Vickers indentation. (adapted from Harding et al. 1994)

When $\frac{c}{a} \gg 1$, the classic Lawn–Evans–Marshall (LEM) model (Lawn et al. 1980) can be adopted

$$K_c = \alpha \cdot \sqrt{\frac{E}{H}} \frac{P_{max}}{c^{3/2}} \tag{5}$$

where α is around 0.016 for Vickers 4-side pyramidal indenter (Liu 2015; Sebastiani et al. 2015a; Ast et al. 2019) and 0.032 for cube- corner indenter (Liu 2015; Xu et al. 2018). For more information about α, see details in Sebastiani et al. (2015a) and Ast et al. (2019). In order to address the issue of the cracking threshold for brittle materials, a cube-corner indenter (normally considered as the sharpest indenter (Pharr 1998)) is normally adopted instead of the standard Berkovich indenter as cracking thresholds can be substantially reduced by employing sharper indenters (Pharr 1998; Harding et al. 1994; Mound and Pharr 2019; Harding 1995).

The crack-length-based method has been widely adopted to measure fracture toughness because of the relative ease of testing and sample preparation (Sebastiani et al. 2015a). However, this method suffers from several disadvantages. First, according to Eq. (4), the selection of an appropriate model requires knowledge of the elastic and plastic properties of the material, the indenter geometry and the lengths of the cracks (which is difficult for small lengths). Similarly, in Eq. (5), the LEM coefficient α, is related to $\frac{E}{H}$, Poisson's ratio (v) and indenter geometry (Sebastiani et al. 2015a; Ghidelli et al. 2017). Second, the thickness of the sample is of great importance because of the development of the plastic zone under the indenter; consequently, radial crack development may be enhanced when the sample thickness is small. In addition, the existence of a microstructure also affects the determination of crack length. Last but not least, the residual stress that exists in materials can alter cracking behaviour significantly (Sebastiani et al. 2015a; Ghidelli et al. 2017), which makes this method impractical if the residual stress is not known precisely. Therefore, an alternative method, namely, the energy analysis method is adopted to determine fracture toughness.

3.2 Energy analysis method

Fracture toughness can be calculated with the energy analysis method (Cheng et al. 2002). During the nanoindentation process, the total energy (U) is composed of two parts, namely elastic energy (U_e) and plastic energy (U_p). The plastic energy (U_p) also consists of two parts, namely pure plastic energy (U_{pp}) and induced fracture energy (U_{frac}). Hence,

$$\mathrm{U} = U_e + U_p = U_e + U_{pp} + U_{frac} \tag{6}$$

U and U_e can be calculated according to the nanoindentation load–displacement curve (Fig. 6).

According to linear elastic fracture mechanics, fracture toughness K_{IC} can be calculated for plane stress conditions:

$$K_{IC} = \sqrt{G_c E_r} \tag{7}$$

and for plane strain conditions:

$$K_{IC} = \sqrt{G_c E_r (1 - \upsilon^2)} \tag{8}$$

where G_c is the strain-energy-release rate and E_r is the reduced elastic modulus. G_c can be calculated using the following equation:

$$G_c = \frac{\partial U_{frac}}{\partial A_c} = \frac{U_{frac}}{A_c} \tag{9}$$

where A_c is the projected contact area, and

$$U_{frac} = U_p - U_{pp} = U - U_e - U_{pp} \tag{10}$$

Although total energy U and elastic energy U_e can be obtained from the P–h curve, the unknown term U_{pp}

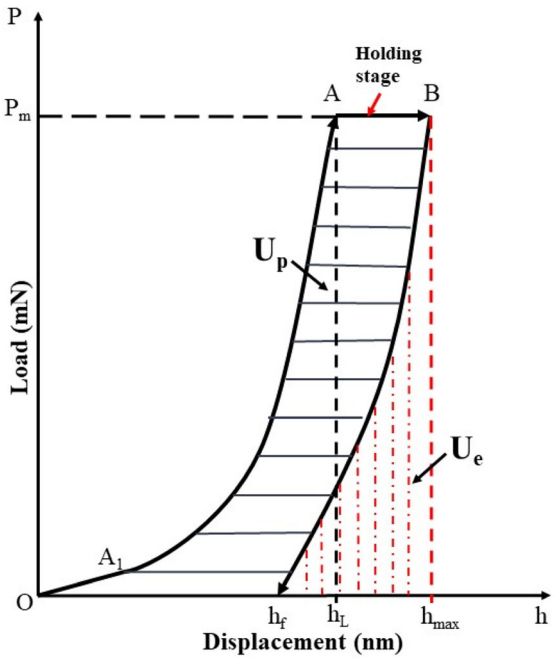


Fig. 6 Schematic illustration of energy analysis method for calculation purposes

cannot be gained directly from the P–h curve and it is normally obtained from combined experimental and finite element modelling (FEM) results.

According to Cheng et al. (2002),

$$U_{pp} = \left\{ 1 - \left[\frac{1 - 3\left(\frac{h_f}{h_{max}}\right)^2 + 2\left(\frac{h_f}{h_{max}}\right)^3}{1 - \left(\frac{h_f}{h_{max}}\right)^2} \right] \right\} U \qquad (11)$$

Therefore, fracture toughness can be obtained by substituting Eq. (11) into (9) and (10). During this process, the crack length does not need to be known. Some representative articles can be referred to when calculating fracture toughness using the energy analysis method. For instance, Zeng et al. (2017) pointed out that Liu et al. (2016) study neglected the maximum holding stage in the indentation-displacement curves. Later, Su et al. (2019) compared Liu's and Zeng et al.'s methods and proposed a revised method. In Su et al.'s model, the irreversible elastic energy by energy area was measured and the impact of holding stage is taking into consideration. It was found that the calculated fracture toughness value using Su et al.'s is in the middle of the range gained from Liu's and Zeng et al.'s methods.

3.3 Micro-cantilever method

With the help of focused ion beam (FIB) milling or lithography, various specimen geometries can be produced to investigate fracture toughness using the nanoindentation method via applied force and measured displacements (Wheeler et al. 2015). The specimen geometries can be single cantilever beams, champed beams, double cantilever beams, and pillars (Sebastiani et al. 2015a). The micro-cantilever method is normally adopted to analyse the fracture mechanisms in brittle and semi-brittle materials (Ast et al. 2019), and two common specimen geometries are the single-cantilever beam and the double-cantilever beam. For the single-cantilever beam, fracture toughness can be calculated using the following equation (Sebastiani et al. 2015a):

$$K_{IC} = \sigma_c \cdot \sqrt{\pi a} \cdot F\left(\frac{a}{b}\right) \qquad (12)$$

where σ_c is the fracture stress, a is the crack length and $F\left(\frac{a}{b}\right)$ is a dimensionless shape factor that depends on the sample geometry.

For the double-cantilever beam, fracture toughness can be calculated using the following equation (Sebastiani et al. 2015a):

$$K_{IC} = \sqrt{3}\frac{(e - \mu h)}{l d^{3/2}} P_c \qquad (13)$$

where e is the distance between the line of action of the load P and the neutral axis of the beam, μ is the friction coefficient between the flat punch indenter, h is the distance from the top of the sample to the tip of the crack, l is the width of the specimen and d is the half width of the sample, P_c is the critical load for crack extension.

The cantilevers must be milled at the edge of the specimen in order to perform bending for the single cantilever beam technique, which is the main disadvantage of this method. For double-cantilever beams, however, a tedious calibration procedure is needed to estimate the friction coefficient between the indenter and the specimen. For both methods, sample preparation may be affected by FIB milling damage (Sebastiani et al. 2015a; Ghidelli et al. 2017). In order to avoid these issues, the micro-pillar splitting method has been developed.

3.4 Micro-pillar splitting method

The micro-pillar splitting method was developed by Sebastiani et al. (2015a, b), and this method is particularly advantageous because it eliminates three major drawbacks of the above methods. First, the indentation crack length and geometry do not need to be accurately determined. Second, the film stress on the measurement can be eliminated. Last but not least, FIB damage can be negligible because crack nucleation and growth before instability occurs in the core of the micro-pillar (Sebastiani et al. 2015a; Ast et al. 2019; Ghidelli et al. 2017). Fracture toughness can be determined by

$$K_{IC} = \gamma \frac{P_c}{R^{3/2}} \tag{14}$$

where P_c is the critical splitting load, R is the pillar radius and γ is the calibration coefficient, which is material-specific. Therefore, the main drawback of this method is that as the calibration coefficient γ is unknown, it must be calculated using numerical techniques, normally cohesive zone (CZ) finite element analysis (Bruns et al. 2020).

4 Time-dependent creep

Since reservoir engineering normally lasts for several decades or even longer; understanding the time-dependent creep properties of geomaterials is of vital importance to high-level radioactive waste (HLRW) repositories stability, wellbore stability, proppant embedment, fracture conductivity reduction, tunnels, mines and reservoir subsidence (Mighani et al. 2015; Shi et al. 2020b). Creep happens due to the viscoelastic properties (also known as the rheological properties) of rocks (Ostadhassan et al. 2018; Liu et al. 2019b) and the correct measurement of the creep of rock has a significant impact on the strength, long-term deformation and seepage characteristics (Shi et al. 2020b). Creep response of a material is closely related to the extrinsic and intrinsic properties, such as temperature, loading rate, creep time, peak load, elastic properties of materials and average grain size of polycrystalline materials (Gan and Tomar 2011). Nanoindentation methods have been adopted to analyse the viscoelastic behaviour of materials by measuring creep depth as a function of time (Liu et al. 2018d). Nanoindentation tests can be carried out to uncover the microphysical deformation mechanisms of geomaterials and it can be utilized as an additional and complementary method (Thom 2019).

Phani et al. (2017) reviewed various methods to measure the indentation creep response and they concluded that the advantages and disadvantages of these methods, namely, constant load and hold test (CLH) (first proposed by Mayo et al. 1990), constant strain rate test (Lucas 1997; Lucas and Oliver 1999), strain rate jump test (Alkorta et al. 2008; Maier et al. 2011) and constant rate of loading test (Mayo and Nix 1988). It was concluded that short duration CLH tests with fast response displacement sensors are superior to other methods (Goodall and Clyne 2006) because this method can be utilized to investigate the strain rates dependence in a single experiment and to minimize the impact of a depth dependent hardness and thermal drift, which is in agreement with Peykov et al. (2012). Apart from the above four common methods, constant contact pressure (Prach et al. 2019) indentation creep method was also proposed to measure the indentation creep.

According to Heap et al. (2009) and Mighani et al. (2015), the creep of rocks has three stages: (1) primary or decelerating stage; (2) secondary or stationary stage; (3) tertiary or accelerating stage. However, depending on the rock types and test conditions, three regimes may not be observed in nanoindentation creep tests (Goodall and Clyne 2006; Liu et al. 2019b). For instance, Gupta et al. (2018a) found that nanoindentation creep of shale composes of only two regions, namely primary creep region and secondary creep, which differs from the conventional creep tests in the laboratory because of the complexed stress underneath the indenter, which is in agreement with Gan and Tomar (2011) (Fig. 7). Mighani et al. (2019) found that displacements and strains approximately depended on the logarithm of time (also known as log time creep) for both nanoindentation creep tests, which is in agreement with Slim (2016), Vandamme et al. (2012) and Vandamme and Ulm (2013), as shown in the following equation

$$\mathrm{h(t)} - h_0 \approx x_1 \ln(t - t_0 + 1) \tag{15}$$

where h is the indentation depth, x_1 a fitting constant, and h_0 and t_0 the depth and time at the beginning of the holding stage. In general, nanoindentation creep tests are represented by power law function. Mighani et al.

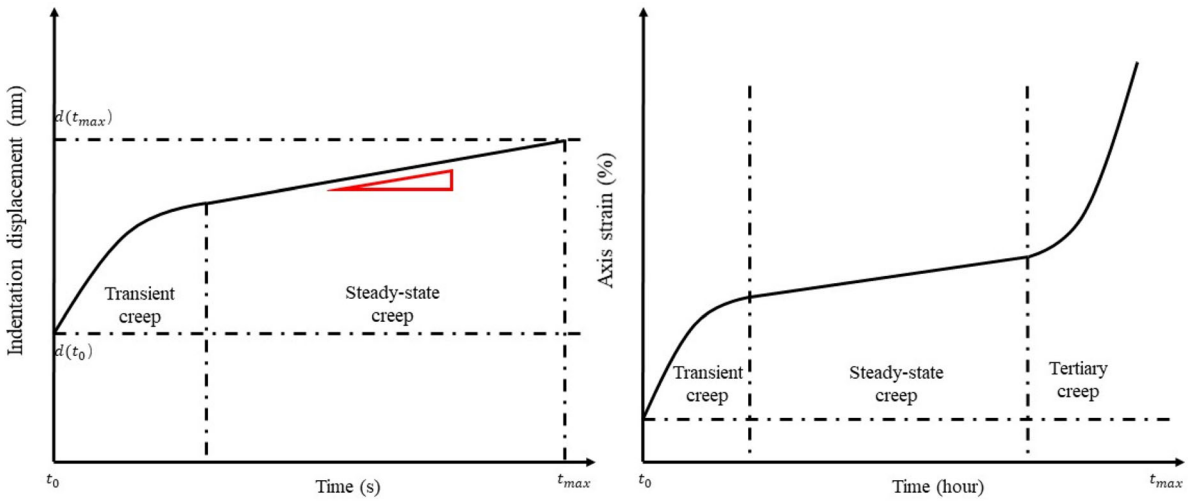


Fig. 7 Comparison between lab-scale creep tests and nanoindentation creep test

(2019) uncovered that when the stress exponent value n is very small, say smaller than 0.1, both the logarithmic and power law model can be utilized to fit the creep data and both models yield nearly the same strain rate decays, namely, $1/t$ and $1/t^{1-n}$.

Previous studies have found that the creep response of samples in nanoindentation tests could be a reliable representation of macro-scale creep behaviour (Liu et al. 2018d; Mighani et al. 2019). For instance, Takagi and Fujiwara (2014) found that the uniaxial creep data can be extracted from the indentation data using the conversion coefficients. However, Goodall and Clyne (2006) contended that there is no correlation between the values measured by nanoindentation and the values from conventional creep testing because the material is under the primary creep regime instead of steady-state creep. Similarly, Choi et al. (2012) found that the stress exponent (n) values are of great difference and they argued that the main reasons for this is due to the fundamental difference between the indentation creep and the conventional uniaxial creep tests (namely, large discrepancy in the stress state between two methods) as well as to the sharp tip geometry. It is suggested that spherical indentation creep and pillar compression creep tests to be carried out to better understand the small-scale creep properties.

Thermal drift error can influence the validity of long-term nanoindentation creep tests and this issue can be solved by using the nano-Dynamic Mechanical Analysis (nano-DMA) method because the modulus is derived at the beginning of the dwelling period (Liu et al. 2018d). This method measures mean contact pressure, penetration depth and dynamic contact stiffness continuously (Mighani et al. 2018a; Liu et al. 2018d, 2019a). In nano-DMA creep tests, the quasi-static force is kept constant while a small oscillation is superimposed in order to measure contact stiffness continuously (Liu et al. 2018d, 2019a) and it can also be utilized for modulus mapping (Wilkinson 2014).

Gupta et al. (2018a) found that creep increases with weak matrix components (i.e., TOC + clays + porosity) and decreases as a function of strong-matrix components (quartz + carbonates) in shale samples, which is in agreement with Slim et al. (2018). Similarly, Liu et al. (2018c) found that samples with higher hardness values showed less creep deformation for the same creep time. They also found that the E, H and K_{IC} of shale samples decreased with increased creep time. Apart from the constituent minerals and creep time, the influence of water and/or oil content on creep deformation has also been studied (Mighani et al. 2018a). It was found that creep deformation increased for water-saturated carbonate compared with dry indentation while the creep magnitude of oil-saturated carbonate was less than that of dry samples. Sharma et al. (2019) investigated the impact of temperature on the creep properties of shales and they found that the increase of temperature causes

the removal of clay-bound water when temperature above 200 °C, which influenced the time-dependent deformation because of the mechanical compaction and grain sliding due to the help of water content.

During the steady-state stage, the creep rate $\dot{\varepsilon}$ are normally correlated to the uniaxial creep stress σ through the power law equation:

$$\dot{\varepsilon} = \alpha\sigma^n \tag{16}$$

where α is the creep coefficient and n is the stress or creep exponent. Similarly, the indentation strain rate $\dot{\varepsilon}_i$ is correlated to the indentation contact pressure p through the following equation according to the Bower et al. (1993)

$$\dot{\varepsilon}_i = Ap^n \exp\left(-\frac{Q}{RT}\right) \tag{17}$$

where $\dot{\varepsilon}_i$ is the nanoindentation creep strain rate, A is a constant, Q is the activation energy, R is the universal gas constant, T is the temperature and n is the power law (creep) exponent (which equals to 1/m, m is the creep strain rate sensitivity). Normally, creep tests were carried out at room temperature, namely, T is a constant value. Hence, the above equation can be simplified as (Mayo and Nix 1988)

$$\dot{\varepsilon}_i = \left(\frac{1}{h}\right)\left(\frac{dh}{dt}\right) = \beta p^n \tag{18}$$

where $\frac{dh}{dt}$ is the indentation displacement rate (indenter velocity), β is the indentation creep coefficient and n is the stress or creep exponent. Therefore, it is of great significance to relate the uniaxial power-law creep properties to the nanoindentation tests (Su et al. 2013) (Fig. 8).

The stress exponent n value can be utilized to uncover the creep mechanism of materials (Gan and Tomar 2011), as shown in the following Table 3.

Thom et al. (2018) investigated the impact of relative humidity (adsorbed water) on the creep rate and hardness (yield strength) of quartz and they found that the deformation of quartz is independent of relative humidity, which indicated that the contact quantity hypothesis was not correct and alternative mechanism of friction aging should be proposed.

Indentation creep test can also be carried out to probe thermally activated process and uncover the underlying thermally activated deformation mechanisms through using two important parameters, namely, strain-rate sensitivity m or the corresponding activation volume V^* (Maier-Kiener and Durst 2017). Chinh and Szommer (2014) analysed mathematically the creep behavior of holding stage of nanoindentation tests and the strain rate sensitivity is obtained through the corresponding indentation depth–time relationship.

5 Drawbacks

5.1 Modulus mapping

Some researchers believe that the nanoindentation test is a destructive technique since it can only provide micro-scale mechanical properties instead of true nanometer-scale resolution. Hence, a novel atomic force microscopy technique (Peak Force QNM^{TM}) needs to be adopted for modulus mapping at the nanometer scale (Eliyahu et al. 2015; Ostadhassan et al. 2018; Li et al. 2017).

5.2 In-situ nanoindentation

Normally, nanoindentation tests are carried out at a fairly narrow window of temperature (e.g. 20–25 °C), humidity (RH $\sim$ 40–50%) and strain rate ($\sim$ 10^{-1} to 10^{-3} s^{-}) (Beake et al. 2016). However, geomaterials are sensitive to temperature changes and apart from ex situ nanoindentation test, in situ nanoindentation tests have also been performed at high- (Wheeler et al. 2015; Duan and Hodge 2009; Wheeler and Michler 2013; Monclús et al. 2014) or low-temperatures (Wang et al. 2019b). High-temperature nanoindentation can be utilized to investigate the temperature-dependent mechanical properties, deformation mechanisms, rheology, and phase changes (Packard et al. 2016). Compared to the room-temperature nanoindentation, high-temperature nanoindentation needs to solve the following issues, such as thermal drift (Schuh et al. 2006), indenter loss or failure (Wheeler and Michler 2013) and sample oxidation (Packard et al. 2016) in order to avoid the influence on the correctness of measured mechanical properties.

Fracture onset and crack propagation, viscoelastic behaviour, crystallographic and compositional information can be obtained from in situ nanoindentation using SEM (Nowak et al. 2010; Huang et al. 2012; Huang and Zhao 2014) or TEM. In-situ SEM and TEM

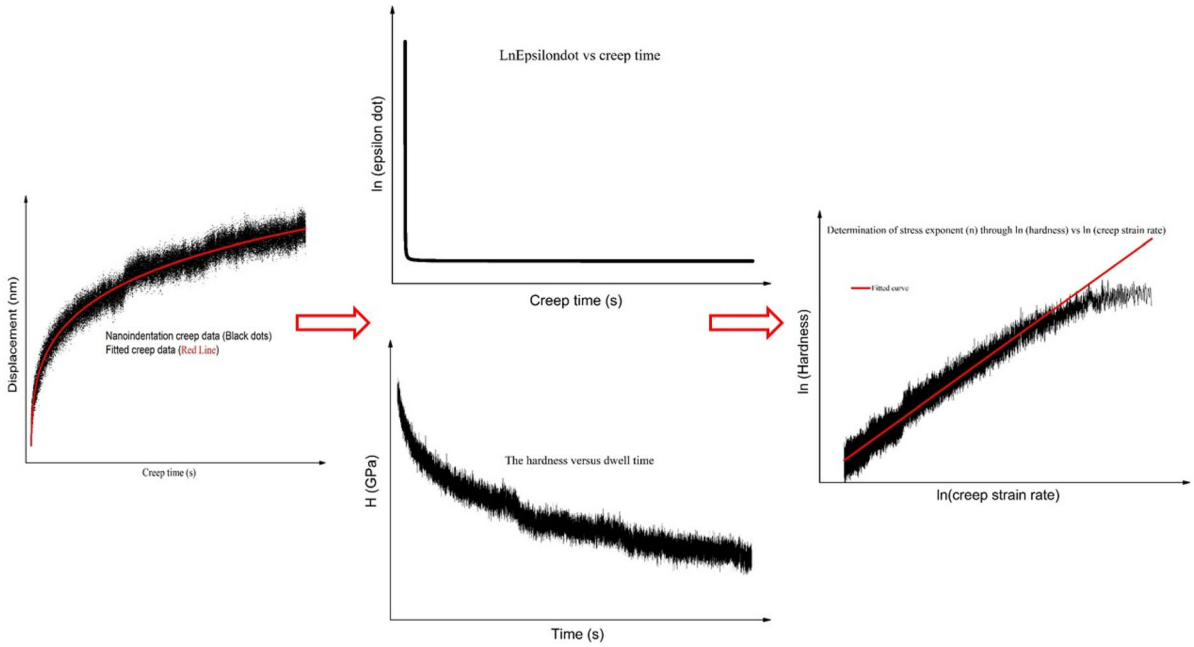


Fig. 8 Steps to calculate the stress exponent using the depth–time plots

Table 3 Stress exponent n value versus its corresponding creep mechanism

Stress exponent value	Corresponding creep mechanism
$n = 1$	Coble creep or Nabarro–Herring creep
$n = 2$	Grain boundary sliding
$n = 3$	Diffusion-controlled dislocation movement

indentation tests can give an in-depth understanding of elastic/plastic deformation behavior of indented materials (Bhushan 2017). Moreover, AFM can also be utilized together with nanoindentation system to provide information about nanoindentation deformation since it is suitable for imaging nanometer-scale indents (Bhushan 2017) and elastic moduli can be obtained from the force-indentation data (Nguyen and Chung 2019). AFM-based nanoindentation method is especially suitable for ultra-thin films or near-surface regions of materials due to its super-high load-resolving capability and small size of indenter (Su and Li 2014), refer to Table 4. In addition, acoustic emission (AE) can also be utilized together with nanoindentation since this technique can be utilized to uncover the plastic deformation, nucleation and growth of cracks of materials (Faisal et al. 2011).

More in situ nanoindentation tests need to be performed to investigate the impacts of temperature, moisture and other environmental parameters on the mechanical parameters of geomaterials, because most geomaterials are subjected to very harsh environmental conditions.

5.3 Cyclic loading

Currently, most nanoindentation tests are carried out under quasi-static conditions. Geomaterials are normally subjected to cyclic loading, which can cause brittle materials to fail at a stress levels lower than their strength under monotonic conditions and this phenomenon is known as “fatigue” (Xiao et al. 2008). The main reason for this phenomenon is due to the fact that the subsequent indentations occurs on a pre-damaged samples after the monotonic loading and further plastic deformation will occur during the cyclic loading (Cui et al. 2018). For instance, Toyama et al. (Toyama et al. 2018) found that the loading level of concrete is below the critical failure load of monotonic loading due to the sub-critical crack growth along with

Table 4 Comparison between nanoindentation and AFM (adapted from Cohen and Kalfon-Cohen 2013)

Characteristic	Nanoindentation	AFM
Load sensitivity	10 nN	< 0.05 nN
Displacement sensitivity	1 nm	0.05 nm or better
nm-scale imaging	Non-existent to fair	Excellent

the cyclic loading. Yin and Zhang (2011a, b) carried out cyclic nanoindentation tests on muscovite and they found that the existence of critical maximum load is vitally important because it impacts the material's responses under different cyclic loads. The occurrence of the shakedown phenomenon influences the measured elastic modulus. Skrzypczak et al. (2009) investigated the ductile–brittle transition of $CaCO_3$ polycrystalline samples using the nanoindentation fatigue test and they found that the transfer from brittle to ductile closely related to the indentation cycles. More nanoindentation tests with cyclic loading should be carried out to investigate changes in mechanical properties under cyclic loading.

5.4 Roughness

Before nanoindentation testing, samples need to be polished because roughness of the sample surface may affect the accuracy of the results (Yang et al. 2017; Wang and Zhang 2018), and the surface roughness should be less than five times the depth of the nanoindentations (Miller et al. 2008). For metallic materials, it is recommended that the indentation depth should be 20 times greater than the Ra surface roughness (Beake et al. 2016; Campbell et al. 2019). Donnelly et al. (2006) found that for the nanoindentation of bones, the indentation depths should be larger than 3 times of root mean squared (RMS) roughness in order to avoid the impact of roughness. Normally, the impact of surface roughness can be reduced through increasing the penetration depth as long as the penetration depth is smaller than the characteristic size of the phases that we are interested (Li and Sakhaee-Pour 2016).

Geomaterials are hard to polish because of the mismatch in properties between their constituent phases (Mahabadi et al. 2012; Randall et al. 2009). It was found that larger surface roughness would increase the scatter of the measured mechanical properties as well as an overall reduction in these properties (Bobko 2008). A large number of nanoindentation tests need to be performed because high surface roughness variations can induce large statistical deviations in indentation results (Randall et al. 2009). Alternatively, higher loads can be applied to overcome this problem (Ghorbal et al. 2017).

Furthermore, during the sample preparation process, residual stress and plastic strain induced by the polishing process can also affect the mechanical properties of geomaterials (Li et al. 2015). Since residual stresses can influence nanoindentation tests significantly, nanoindentation tests needs to be carried out to determine the residual stresses (Xu and Li 2008; Sebastiani et al. 2011). The impacts of other factors, such as indentation space (Hintsala et al. 2018; Sudharshan Phani and Oliver 2019), tip wear (indenter loss) (Hintsala et al. 2018), sample tilt (Xu and Li 2007; Kashani and Madhavan 2011; Wang and Liu 2019) and contaminant-geomaterials interactions can also be investigated to study the mechanical properties of geomaterials in nanoindentation tests.

6 Conclusions

Nanoindentation tests have been carried out to study the mechanical parameters of various geomaterials, including Young's modulus, hardness, fracture toughness, time-dependent creep properties and tensile strength. In this article, we have reviewed the application of nanoindentation tests to various rocks, such as shale, coal, limestone, marble, sandstone and claystone.

In addition, we have discussed various methods to calculate the fracture toughness of geomaterials, including the crack-length-based method, the energy analysis method, the micro-cantilever method and the micro-pillar splitting method.

Although nanoindentation techniques have many advantages over traditional methods to obtain the mechanical properties of geomaterials, they have some disadvantages.

(1) In-situ nanoindentation is one of the main foci for future studies because external

environmental conditions affect the mechanical properties of geomaterials significantly. The impacts of high-temperature, low-temperature, moisture content and environmental factors on the mechanical properties of geomaterials should be studied. Various loading condition (cyclic loading, impact loading) can also be adopted to study the mechanical parameters variations.

(2) In addition, the combination of in situ SEM or TEM and nano-indenters should be adopted to study the elastic/plastic deformation behaviour of geomaterials. Electron backscatter diffraction (EBSD) can also be utilized to investigate the impacts of various factors, such as crystal orientation, texture, etc. on the mechanical parameters change. In addition, dislocation density can be gained using EBSD and this parameter can be helpful to uncover the deformation mechanism of rock minerals from a microscale perspective.

(3) Last but not least, the inverse analysis should be carried out to obtain the mechanical properties (elasto-plastic, visco-plastic, etc.) of constitutive model of geomaterials.

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References

Abedi S, Slim M, Hofmann R, Bryndzia T, Ulm F-J (2016a) Nanochemo-mechanical signature of organic-rich shales: a coupled indentation–EDX analysis. Acta Geotech 11(3):559–572

Abedi S, Slim M, Ulm F-J (2016b) Nanomechanics of organic-rich shales: the role of thermal maturity and organic matter content on texture. Acta Geotech 11(4):775–787

Abousleiman Y, Tran M, Hoang S, Ortega JA, Ulm F-J (2010) Geomechanics field characterization of Woodford Shale and Barnett Shale with advanced logging tools and nano-indentation on drill cuttings. Lead Edge 29(6):730–736

Abousleiman Y, Hull K, Han Y, Al-Muntasheri G, Hosemann P, Parker S, Howard C (2016) The granular and polymer composite nature of kerogen-rich shale. Acta Geotech 11(3):573–594

Ahmadov R, Vanorio T, Mavko G (2009) Confocal laser scanning and atomic-force microscopy in estimation of elastic properties of the organic-rich Bazhenov Formation. Lead Edge 28(1):18–23

Akono A-T, Kabir P (2016) Nano-scale characterization of organic-rich shale via indentation methods. In: Jin C, Cusatis G (eds) New frontiers in oil and gas exploration. Springer, Cham, pp 209–233

Akono A-T, Kabir P, Shi Z, Fuchs S, Tsotsis TT, Jessen K, Werth CJ (2019) Modeling CO_2-induced alterations in Mt. Simon Sandstone via nanomechanics. Rock Mech Rock Eng 52(5):1353–1375

Akrad OM, Miskimins JL, Prasad M (2011) The effects of fracturing fluids on shale rock mechanical properties and proppant embedment. In: SPE annual technical conference and exhibition. Society of Petroleum Engineers

Alcalá J, Esque-De Los Ojos D, Rodríguez S (2009) The role of crystalline anisotropy in mechanical property extractions through Berkovich indentation. J Mater Res 24(3):1235–1244

Alkorta J, Martínez-Esnaola JM, Sevillano JG (2008) Critical examination of strain-rate sensitivity measurement by nanoindentation methods: application to severely deformed niobium. Acta Mater 56(4):884–893

Alsinan S (2017) Effect of artificially induced maturation on the elastic properties of kerogen. SEG technical program expanded abstracts 2017, Society of exploration geophysicists, pp 3647–3652

Alstadt KN, Katti KS, Katti DR (2016) Nanoscale morphology of kerogen and in situ nanomechanical properties of green river oil shale. J Nanomech Micromech 6(1):04015003

Ante M, Lingareddy M, Aminzadeh F, Jha B (2018) Nano-and micro-scale deformation behavior of sandstone and shale. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Arnold G, Auvray C, Giraud A, Armand G (2015) Claystone phases mechanical properties identified from temperature and humidity controlled nanoindentation measurements. In: 13th ISRM international congress of rock mechanics. International Society for Rock Mechanics and Rock Engineering

Ashraf W, Olek J, Tian N (2015) Nanomechanical characterization of the carbonated wollastonite system. In: Sobolev K, Shah SP (eds) Nanotechnology in construction. Springer, Cham, pp 71–77

Ast J, Ghidelli M, Durst K, Goeken M, Sebastiani M, Korsunsky A (2019) A review of experimental approaches to fracture toughness evaluation at the micro-scale. Mater Des 173:107762

Auvray C, Giot R, Grgic D (2013a) Nano-indentation partially saturated argillite: experience device and measurements. In: Proceedings of EUROCK

Auvray C, Giot R, Grgic D (2013b) Nano-indentation tests on partially satured argillite: experimental device and measurements. In: Kwasniewski M, Lydzba D (eds) Rapid-Miner: data mining use cases and business analytics applications, Cambridge, p 201

Auvray C, Arnold G, Armand G (2015) Experimental study of elastic properties of different constituents of partially

saturated argillite using nano-indentation tests. Eng Geol 191:61–70

Auvray C, Lafrance N, Bartier D (2017) Elastic modulus of claystone evaluated by nano-/micro-indentation tests and meso-compression tests. J Rock Mech Geotech Eng 9(1):84–91

Ban H, Karki P, Kim Y-R (2014) Nanoindentation test integrated with numerical simulation to characterize mechanical properties of rock materials. J Test Eval 42(3):787–796

Bandini A, Berry P, Bemporad E, Sebastiani M (2012) Effects of intra-crystalline microcracks on the mechanical behavior of a marble under indentation. Int J Rock Mech Min Sci 54:47–55

Bandini A, Berry P, Bemporad E, Sebastiani M, Chicot D (2014) Role of grain boundaries and micro-defects on the mechanical response of a crystalline rock at multiscale. Int J Rock Mech Min Sci 71:429–441

Bartier D, Auvray C (2017) Determination of elastic modulus of claystone: nano-/micro-indentation and meso-compression tests used to investigate impact of alkaline fluid propagation over 18 years. J Rock Mech Geotech Eng 9(3):511–518

Bastos de Paula O (2011) Elastic properties of carbonates: measurements and modelling. Department of Exploration Geophysics, Curtin University, Bentley, p 149

Basu S, Moseson A, Barsoum MW (2006) On the determination of spherical nanoindentation stress–strain curves. J Mater Res 21(10):2628–2637

Basu S, Zhou A, Barsoum M (2009) On spherical nanoindentations, kinking nonlinear elasticity of mica single crystals and their geological implications. J Struct Geol 31(8):791–801

Beake BD, Harris AJ, Liskiewicz TW (2016) Advanced nanomechanical test techniques. In: Ranganathan NM (ed) Materials characterization: modern methods and applications. CRC Press, Boca Raton, pp 17–18

Beirau T, Nix WD, Ewing RC, Schneider GA, Groat LA, Bismayer U (2016) Mechanical properties of natural radiation-damaged titanite and temperature-induced structural reorganization: a nanoindentation and Raman spectroscopic study. Am Miner 101(2):371–384

Bennett KC, Berla LA, Nix WD, Borja RI (2015) Instrumented nanoindentation and 3D mechanistic modeling of a shale at multiple scales. Acta Geotech 10(1):1–14

Berthonneau J, Hoover CG, Grauby O, Baronnet A, Pellenq RJ-M, Ulm F-J (2017) Crystal-chemistry control of the mechanical properties of 2: 1 clay minerals. Appl Clay Sci 143:387–398

Bhushan B (2017) Depth-sensing nanoindentation measurement techniques and applications. Microsyst Technol 23(5):1595–1649

Bobko C (2006) Material invariant properties and reconstruction of microstructure of sandstones by nanoindentation and microporoelastic analysis. Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, Cambridge, p 189

Bobko C (2008) Assessing the mechanical microstructure of shale by nanoindentation: the link between mineral composition and mechanical properties. Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, Cambridge, p 351

Bobko C, Ulm F-J (2008) The nano-mechanical morphology of shale. Mech Mater 40(4):318–337

Borodich FM, Bull S, Epshtein S (2015) Nanoindentation in studying mechanical properties of heterogeneous materials. J Min Sci 51(3):470–476

Boulenouar A, Mighani S, Pourpak H, Bernabé Y, Evans B (2017) Mechanical properties of Vaca Muerta shales from nano-indentation tests. In: 51st US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Bower A, Fleck NA, Needleman A, Ogbonna N (1993) Indentation of a power law creeping solid. Proc R Soc Lond Ser A Math Phys Sci 441(1911):97–124

Brooks Z (2010) A nanomechanical investigation of the crack tip process zone of marble. Civil and Environmental Engineering, Massachusetts Institute of Technology, Cambridge

Brooks Z, Ulm F-J, Einstein H, Abousleiman Y (2010) A nanomechanical investigation of the crack tip process zone. In: 44th US rock mechanics symposium and 5th US-Canada rock mechanics symposium. American Rock Mechanics Association, Salt Lake City, UT

Brooks Z, Ulm F-J, Einstein H (2012) Role of microstructure size in fracture process zone development of marble. In: 46th US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Brooks Z, Ulm F-J, Einstein H (2013) Environmental scanning electron microscopy (ESEM) and nanoindentation investigation of the crack tip process zone in marble. Acta Geotech 8(3):223–245

Broz ME, Cook RF, Whitney DL (2006) Microhardness, toughness, and modulus of Mohs scale minerals. Am Miner 91(1):135–142

Bruns S, Petho L, Minnert C, Michler J, Durst K (2020) Fracture toughness determination of fused silica by cube corner indentation cracking and pillar splitting. Mater Des 186:108311

Cała M, Cyran K, Kawa M, Kolano M, Łydżba D, Pachnicz M, Rajczakowska M, Różański A, Sobótka M, Stefaniuk D (2017) Identification of microstructural properties of shale by combined use of X-ray micro-CT and nanoindentation tests. Procedia Eng 191:735–743

Campbell AC, Buršíková V, Martinek J, Klapetek P (2019) Modeling the influence of roughness on nanoindentation data using finite element analysis. Int J Mech Sci 161:105015

Cheng Y-T, Cheng C-M (2004) Scaling, dimensional analysis, and indentation measurements. Mater Sci Eng R Rep 44(4–5):91–149

Cheng Y-T, Li Z, Cheng C-M (2002) Scaling relationships for indentation measurements. Philos Mag A 82(10):1821–1829

Chinh NQ, Szommer P (2014) Mathematical description of indentation creep and its application for the determination of strain rate sensitivity. Mater Sci Eng, A 611:333–336

Choi I-C, Yoo B-G, Kim Y-J, Jang J-I (2012) Indentation creep revisited. J Mater Res 27(1):3–11

Cohen SR, Kalfon-Cohen E (2013) Dynamic nanoindentation by instrumented nanoindentation and force microscopy: a comparative review. Beilstein J Nanotechnol 4(1):815–833

Constantinides G, Ulm F-J, Van Vliet K (2003) On the use of nanoindentation for cementitious materials. Mater Struct 36(3):191–196

Constantinides G, Ravi Chandran KS, Ulm FJ, Van Vliet KJ (2006) Grid indentation analysis of composite microstructure and mechanics: principles and validation. Mater Sci Eng, A 430(1):189–202

Corapcioglu H, Miskimins J, Prasad M (2014) Fracturing fluid effects on Young's modulus and embedment in the niobrara formation. In: SPE annual technical conference and exhibition. Society of Petroleum Engineers, Amsterdam, The Netherlands, p 17

Cui Y-Y, Jia Y-F, Xuan F-Z (2018) Micro-deformation evolutions of the constituent phases in duplex stainless steel during cyclic nanoindentation. Sci Rep 8(1):6199

Daphalapurkar N, Wang F, Fu B, Lu H, Komanduri R (2011) Determination of mechanical properties of sand grains by nanoindentation. Exp Mech 51(5):719–728

Deirieh A, Ortega JA, Ulm FJ, Abousleiman Y (2012) Nanochemomechanical assessment of shale: a coupled WDS-indentation analysis. Acta Geotech 7(4):271–295

Deng P (2017) Measurement of mineral elastic modulus through nanoindentation. Auburn University, Auburn

Donnelly E, Baker SP, Boskey AL, van der Meulen MC (2006) Effects of surface roughness and maximum load on the mechanical properties of cancellous bone measured by nanoindentation. J Biomed Mater Res, Part A 77(2):426–435

Du H, Radonjic M (2018) Comparison of micro/nano-indentation results for Pottsville and Marcellus Shale. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Du J, Hu L, Meegoda JN, Zhang G (2018) Shale softening: observations, phenomenological behavior, and mechanisms. Appl Clay Sci 161:290–300

Duan ZC, Hodge AM (2009) High-temperature nanoindentation: new developments and ongoing challenges. JOM 61(12):32

Dutta A, Penumadu D (2007) Hardness and modulus of individual sand particles using nanoindentation. In: Geo-denver 2007 congress: new peaks in geotechnics. pp 1–10

Eliyahu M, Emmanuel S, Day-Stirrat RJ, Macaulay CI (2015) Mechanical properties of organic matter in shales mapped at the nanometer scale. Mar Pet Geol 59:294–304

Epshtein SA, Borodich FM, Bull SJ (2015) Evaluation of elastic modulus and hardness of highly inhomogeneous materials by nanoindentation. Appl Phys A 119(1):325–335

Erdoğan S, Forster A, Stutzman P, Garboczi E (2017) Particle-based characterization of Ottawa sand: shape, size, mineralogy, and elastic moduli. Cement Concr Compos 83:36–44

Faisal N, Ahmed R, Reuben R (2011) Indentation testing and its acoustic emission response: applications and emerging trends. Int Mater Rev 56(2):98–142

Fan M, Jin Y, Chen M, Geng Z (2019) Mechanical characterization of shale through instrumented indentation test. J Pet Sci Eng 174:607–616

Fang L (2008) Development and applications of electrical instrumented indentation. Dissertation, The Pennsylvania State University

Feng Y, Zhang T (2015) Determination of fracture toughness of brittle materials by indentation. Acta Mech Solida Sin 28(3):221–234

Fischer-Cripps A (2011) Nanoindentation. Springer, New York

Fu H (2018) The application of nanoindentation technology with simulation on the micromechanical properties of bakken formation. Petroleum Engineering, University of North Dakota, Grand Forks, North Dakota

Fu H, Hou X, Ge J, Pu H (2017) Nanoindentation studies on the mechanical properties of bakken formation. In: 51st US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Gan M, Tomar V (2011) Scale and temperature dependent creep modeling and experiments in materials. JOM 63(9):27

Ghidelli M, Sebastiani M, Johanns KE, Pharr GM (2017) Effects of indenter angle on micro-scale fracture toughness measurement by pillar splitting. J Am Ceram Soc 100(12):5731–5738

Ghorbal GB, Tricoteaux A, Thuault A, Louis G, Chicot D (2017) Mechanical characterization of brittle materials using instrumented indentation with Knoop indenter. Mech Mater 108:58–67

Goldsby DL, Rar A, Pharr GM, Tullis TE (2004) Nanoindentation creep of quartz, with implications for rate-and state-variable friction laws relevant to earthquake mechanics. J Mater Res 19(1):357–365

Golovin YI, Tyurin A, Victorov S, Kochanov A, Samodurov A, Pirozhkova T (2017) Physico-mechanical properties and micromechanisms of local deformation in thin near-surface layers of complex multiphase materials. Bull Russ Acad Sci Phys 81(3):360–364

Golovin Y, Tyurin A, Victorov S, Kochanov A, Pirozhkova T (2018) Size effects and charting the physical and mechanical properties of individual phases and interphases in polycrystalline materials. Bull Russ Acad Sci Phys 82(7):856–859

Goodall R, Clyne T (2006) A critical appraisal of the extraction of creep parameters from nanoindentation data obtained at room temperature. Acta Mater 54(20):5489–5499

Goral J, Deo M, Nadimi S, McLannan J, McCarter M, Mattson E, Huang H (2018) Micro-and macro-scale geomechanical testing of Woodford Shale. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Gupta I, Sondergeld CH, Rai CS, Hofmann R (2017) Water weakening: case study from Marcellus and Woodford. In: SPE/AAPG/SEG unconventional resources technology conference, unconventional resources technology conference. Austin, Texas, USA, p 15

Gupta I, Sondergeld C, Rai C (2018a) Applications of nanoindentation for reservoir characterization in shales. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Gupta I, Sondergeld C, Rai C, Hofmann R (2018b) Water weakening: case study from Marcellus, Woodford, Eagle Ford and Wolfcamp. In: 52nd U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, Seattle, Washington, p 9

Gupta I, Rai C, Sondergeld C, Hofmann R (2019) Water weakening: a laboratory study of marcellus. Eagle Ford,

and Wolfcamp Shales, SPE Reservoir Evaluation & Engineering, Woodford

Han Q, Qu Z, Ye Z (2018) Research on the mechanical behaviour of shale based on multiscale analysis. R Soc Open Sci 5(10):181039

Hansen LN, David EC, Brantut N, Wallis D (2020) Insight into the microphysics of antigorite deformation from spherical nanoindentation. Philos Trans R Soc A 378(2165):20190197

Harding DS (1995) Cracking in brittle materials during low-load indentation and its relation to fracture toughness. Dissertation, Rice University

Harding D, Oliver W, Pharr G (1994) Cracking during nanoindentation and its use in the measurement of fracture toughness. MRS Online Proceedings Library Archive 356

Haušild P, Materna A, Kocmanová L, Matějíček J (2016) Determination of the individual phase properties from the measured grid indentation data. J Mater Res 31(22):3538–3548

Haušild P, Materna A, Kocmanová L, Matějíček J (2017) Some issues in relations between microstructure and indentation measurements. In: Šandera P (ed) Solid state phenomena. Trans Tech Publ, pp 131–136

Heap M, Baud P, Meredith P, Bell A, Main I (2009) Time-dependent brittle creep in darley dale sandstone. J Geophys Res Solid Earth 114(B7). https://doi.org/10.1029/2008JB006212

Hintsala ED, Hangen U, Stauffer DD (2018) High-throughput nanoindentation for statistical and spatial property determination. JOM 70(4):494–503

Hou D, Zhang G, Pant R, Shen J, Liu M, Luo H (2016) Nanoindentation characterization of a ternary clay-based composite used in ancient Chinese construction. Materials 9(11):866

Hrbek V, Koudelková V (2017) Grid indentation and statistic deconvolution: limitations and accuracy. In: Ryparová P, Tesárek P (ed) Key engineering materials. Trans Tech Publ, pp 15–22

Hu Z, Du Y, Luo H, Zhong B, Lu H (2014) Internal deformation measurement and force chain characterization of mason sand under confined compression using incremental digital volume correlation. Exp Mech 54(9):1575–1586

Huang H, Zhao H (2014) In Situ nanoindentation and scratch testing inside scanning electron microscopes: opportunities and challenges. Sci Adv Mater 6:875–889

Huang Y, Liu X, Zhou Y, Ma Z, Lu C (2011) Mathematical analysis on the uniqueness of reverse algorithm for measuring elastic-plastic properties by sharp indentation. J Mater Sci Technol 27(7):577–584

Huang H, Zhao H, Mi J, Yang J, Wan S, Xu L, Ma Z (2012) A novel and compact nanoindentation device for in situ nanoindentation tests inside the scanning electron microscope. AIP Adv 2(1):012104

Huen WY, Lee H, Vimonsatit V, Mendis P (2020) Relationship of stiffness-based indentation properties using continuous-stiffness-measurement method. Materials 13(1):97

Hull KL, Abousleiman YN, Han Y, Al-Muntasheri GA, Hosemann P, Parker SS, Howard CB (2017) Nanomechanical characterization of the tensile modulus of rupture for kerogen-rich shale. SPE J 22(04):1024–1033

Jung-Min L, Chan-Joo L, Kyung-Hun L, Byung-Min K (2012) Effects of elastic-plastic properties of materials on residual indentation impressions in nano-indentation using sharp indenter. Trans Nonferr Met Soc China 22:s585–s595

Kabir P, Akono A-T (2018) Fluid-rock reactions in Mt. Simon Sandstone at microscopic length-scale. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Kadali S, Sharma S, Singh D (2013) Application of nanoindentation to establish influence of heat on soils. Eng Geol 162:14–21

Kalidindi SR, Pathak S (2008) Determination of the effective zero-point and the extraction of spherical nanoindentation stress–strain curves. Acta Mater 56(14):3523–3532

Kalo K (2017) Microstructural characterization and micromechanical modeling of oolitic porous rocks. Mechanical-Civil Engineering, University of Lorraine, Nancy

Kang J (2013) Determination of elastic-plastic and visco-plastic material properties from instrumented indentation curves. University of Nottingham, Nottingham

Kashani M, Madhavan V (2011) Analysis and correction of the effect of sample tilt on results of nanoindentation. Acta Mater 59(3):883–895

Keller LM, Schwiedrzik JJ, Gasser P, Michler J (2017) Understanding anisotropic mechanical properties of shales at different length scales: in situ micropillar compression combined with finite element calculations. J Geophys Res Solid Earth 122(8):5945–5955

Komanduri R, Lu H, Subramanian V, Luo H, Wang F, Cooper WL (2011) Mechanics of granular materials: experimentation and simulations for determining the compressive and shear behaviors of sand at granular and meso scales. Oklahoma State University Stillwater School of Mechanical and Aerospace Engineering

Kong L, Ostadhassan M, Fereshtenejad S, Song J-J, Li C (2018) Anisotropy analysis of 3D printed gypsum rocks integrating pulse-transmission, nanoindentation and micro-CT techniques. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Kong L, Ostadhassan M, Zamiran S, Liu B, Li C, Marino GG (2019) Geomechanical upscaling methods: comparison and verification via 3D printing. Energies 12(3):382

Konstantinidis AA, Frantziskonis G, Askes H, Aifantis EC (2016) The use of nanoindentation for determining internal lengths and the constitutive response of monument materials: models and experiments. J Mech Behav Mater 25(1–2):57–60

Kossovich E, Epshtein S, Dobryakova N, Minin M, Gavrilova D (2017) Mechanical properties of thin films of coals by nanoindentation. In: Conference on physical and mathematical modeling of earth and environment processes. Springer, pp 45–50

Kossovich EL, Borodich FM, Epshtein SA, Galanov BA, Minin MG, Prosina VA (2019) Mechanical, structural and scaling properties of coals: depth-sensing indentation studies. Appl Phys A 125(3):195

Koudelka P, Kytýř D, Koudelková V, Lukeš J, Doktor T, Valach J (2014) Material testing of natural stones used in historical buildings based on scanning electron microscopy and nanoindentation. Key Eng Mater 586:186–189

Kranjc K (2017) Characterizing structure, properties, and deformation in metallic glasses and olivine using instrumented nanoindentation. Institute of Materials Science and Engineering, Washington University in St. Louis, p 130

Kranjc K, Rouse Z, Flores KM, Skemer P (2016) Low-temperature plastic rheology of olivine determined by nanoindentation. Geophys Res Lett 43(1):176–184

Kumamoto KM, Thom CA, Wallis D, Hansen LN, Armstrong DE, Warren JM, Goldsby DL, Wilkinson AJ (2017) Size effects resolve discrepancies in 40 years of work on low-temperature plasticity in olivine. Sci Adv 3(9):e1701338

Kumar V, Sondergeld CH, Rai CS (2012a) Nano to macro mechanical characterization of shale. In: SPE annual technical conference and exhibition. Society of Petroleum Engineers

Kumar V, Curtis ME, Gupta N, Sondergeld CH, Rai CS (2012b) Estimation of elastic properties of organic matter in Woodford Shale through nanoindentation measurements. In: SPE Canadian unconventional resources conference. Society of Petroleum Engineers, Calgary, Alberta, Canada, p 11

Kumar V, Sondergeld C, Rai CS (2015) Effect of mineralogy and organic matter on mechanical properties of shale. Interpretation 3(3):SV9–SV15

Lawn BR, Evans A, Marshall D (1980) Elastic/plastic indentation damage in ceramics: the median/radial crack system. J Am Ceram Soc 63(9–10):574–581

Lebedev M, Wilson MEJ, Mikhaltsevitch V (2014a) An experimental study of solid matrix weakening in water-saturated Savonnières limestone. Geophys Prospect 62(6):1253–1265

Lebedev M, Mikhaltsevitch V, Carson M, Pervukhina M, Gurevich B (2014b) Softening of rocks matrix due to water flood: experimental study. SEG Technical Program Expanded Abstracts 2014. Society of Exploration Geophysicists, pp 2760–2765

Lebedev M, Zhang Y, Sarmadivaleh M, Barifcani A, Al-Khdheeawi E, Iglauer S (2017) Carbon geosequestration in limestone: pore-scale dissolution and geomechanical weakening. Int J Greenhouse Gas Control 66:106–119

Li W, Sakhaee-Pour A (2016) Macroscale Young's moduli of shale based on nanoindentations. Petrophysics 57(06):597–603

Li Y, Kanouté P, François M (2015) Disturbance induced by surface preparation on instrumented indentation test. Mater Sci Eng, A 642:381–390

Li C, Ostadhassan M, Kong L (2017) Nanochemo-mechanical characterization of organic shale through AFM and EDS. SEG technical program expanded abstracts 2017, society of exploration geophysicists, pp 3837–3840

Li C, Ostadhassan M, Kong L (2018) Effect of organic matter on nano-mechanical properties of organic-rich shale. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Li X, Liu Z, Cui S, Luo C, Li C, Zhuang Z (2019a) Predicting the effective mechanical property of heterogeneous materials by image based modeling and deep learning. Comput Methods Appl Mech Eng 347:735–753

Li C, Ostadhassan M, Abarghani A, Fogden A, Kong L (2019b) Multi-scale evaluation of mechanical properties of the Bakken shale. J Mater Sci 54(3):2133–2151

Li C, Ostadhassan M, Kong L, Bubach B (2019c) Multi-scale assessment of mechanical properties of organic-rich shales: a coupled nanoindentation, deconvolution analysis, and homogenization method. J Pet Sci Eng 174:80–91

Liu Y (2015) Fracture toughness assessment of shales by nanoindentation. Department of Civil and Environmental Engineering, University of Massachusetts Amherst, Cambridge, p 80

Liu K (2018) Microstructures and nanomechanical properties of the bakken shale. University of North Dakota, Grand Forks, North Dakota, p 238

Liu K, Ostadhassan M (2017) Microstructural and geomechanical analysis of Bakken shale at nanoscale. J Pet Sci Eng 153:133–144

Liu K, Ostadhassan M, Bubach B (2016) Applications of nanoindentation methods to estimate nanoscale mechanical properties of shale reservoir rocks. J Nat Gas Sci Eng 35:1310–1319

Liu K, Ostadhassan M, Li C, Alexeyev A, Hou X (2017a) Fracture toughness measurement of shales using nano-indentation: the bakken case study. In: 51st U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, San Francisco, California, USA, p 6

Liu K, Ostadhassan M, Li C (2017b) Quantifying the nanomechanical signature of shale oil formations by nanoindentation. In: Unconventional resources technology conference society of exploration geophysicists. American Association of Petroleum, Austin, Texas, pp 2075–2083

Liu K, Ostadhassan M, Bubach B, Ling K, Tokhmechi B, Robert D (2018a) Statistical grid nanoindentation analysis to estimate macro-mechanical properties of the Bakken Shale. J Nat Gas Sci Eng 53:181–190

Liu K, Ostadhassan M, Bubach B (2018b) Application of nanoindentation to characterize creep behavior of oil shales. J Pet Sci Eng 167:729–736

Liu K, Ostadhassan M, Wang H (2018c) Creep behavior of shale-nanoindentation experiments. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Liu K, Ostadhassan M, Bubach B, Dietrich R, Rasouli V (2018d) Nano-dynamic mechanical analysis (nano-DMA) of creep behavior of shales: Bakken case study. J Mater Sci 53(6):4417–4432

Liu K, Ostadhassan M, Xu X, Bubach B (2019a) Abnormal behavior during nanoindentation holding stage: characterization and explanation. J Pet Sci Eng 173:733–747

Liu Y, Xiong Y, Liu K, Yang C, Peng P (2019b) Indentation size and loading rate sensitivities on mechanical properties and creep behavior of solid bitumen. Int J Coal Geol 216:103295

Lu Y, Li Y, Wu Y, Luo S, Jin Y, Zhang G (2019) Characterization of shale softening by large volume-based nanoindentation. Rock Mechanics and Rock Engineering, pp 1–17

Lucas BN (1997) An experimental investigation of creep and viscoelastic properties using depth-sensing indentation techniques. University of Tennessee, Knoxville, p 250

Lucas B, Oliver W (1999) Indentation power-law creep of high-purity indium. Metall Mater Trans A 30(3):601–610

Lucca D, Herrmann K, Klopfstein M (2010) Nanoindentation: measuring methods and applications. CIRP Ann 59(2):803–819

Luo H, Du Y, Hu Z, Lu H (2015) High-strain rate compressive behavior of dry Mason sand under confinement. In: Song B, Casem D, Kimberley J (eds) Dynamic behavior of materials, vol 1. Springer, pp 325–333

Ma Y, Huang X, Song Y, Hang W, Yuan J, Zhang T (2019a) Orientation-independent yield stress and activation volume of dislocation nucleation in $LiTaO_3$ single crystal by nanoindentation. Materials 12(17):2799

Ma F, Song J, Luo S, DeGroot DJ, Bai X, Zhang G (2019b) Distinct responses of nanostructured layered muscovite to uniform and nonuniform straining. J Mater Sci 54(2):1077–1098

Ma Z, Gamage RP, Rathnaweera T, Kong L (2019c) Review of application of molecular dynamic simulations in geological high-level radioactive waste disposal. Appl Clay Sci 168:436–449

Magnenet V, Auvray C, Francius G, Giraud A (2011) Determination of the matrix indentation modulus of Meuse/Haute-Marne argillite. Appl Clay Sci 52(3):266–269

Mahabadi O, Randall N, Zong Z, Grasselli G (2012) A novel approach for micro-scale characterization and modeling of geomaterials incorporating actual material heterogeneity. Geophys Res Lett 39(1). https://doi.org/10.1029/2011GL050411

Maier V, Durst K, Mueller J, Backes B, Höppel HW, Göken M (2011) Nanoindentation strain-rate jump tests for determining the local strain-rate sensitivity in nanocrystalline Ni and ultrafine-grained Al. J Mater Res 26(11):1421–1430

Maier-Kiener V, Durst K (2017) Advanced nanoindentation testing for studying strain-rate sensitivity and activation volume. JOM 69(11):2246–2255

Manjunath GL, Jha B (2019) Nanoscale fracture mechanics of Gondwana coal. Int J Coal Geol 204:102–112

Maruvanchery V (2019) Effects of extreme environmental conditions on the mesoscale and nanoscale properties of calcite-cemented sandstone. Colorado School of Mines, Arthur Lakes Library, Golden

Maruvanchery V, Kim E (2020) Mechanical characterization of thermally treated calcite-cemented sandstone using nanoindentation, scanning electron microscopy and automated mineralogy. Int J Rock Mech Min Sci 125:104158

Mashhadian M, Verde A, Sharma P, Abedi S (2018) Assessing mechanical properties of organic matter in shales: Results from coupled nanoindentation/SEM-EDX and micromechanical modeling. J Pet Sci Eng 165:313–324

Mason J, Jordan T, Carloni J, Baker S, Zehnder A (2014) Dependence of micro-mechanical properties on lithofacies: indentation experiments on Marcellus Shale. In: Unconventional resources technology conference, Denver, Colorado, 25–27 August 2014, Society of Exploration Geophysicists, American Association of Petroleum, pp 1758–1770

Mayo M, Nix W (1988) A micro-indentation study of superplasticity in Pb, Sn, and Sn-38 wt% Pb. Acta Metall 36(8):2183–2192

Mayo M, Siegel R, Narayanasamy A, Nix W (1990) Mechanical properties of nanophase TiO_2 as determined by nanoindentation. J Mater Res 5(5):1073–1082

Mba K, Prasad M (2010) Mineralogy and its contribution to anisotropy and kerogen stiffness variations with maturity in the Bakken Shales. SEG technical program expanded abstracts, pp 2612–2616

Mighani S, Taneja S, Sondergeld CH, Rai CS (2015) Nanoindentation creep measurements on shale. In: 49th U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, San Francisco, California, p 7

Mighani S, Bernabe Y, Mok U, Pec M, Evans B (2018a) Can we use nanoindentation to derive the poroelastic P Arameters of microporous rocks?-experimental evidence. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Mighani S, Bernabe Y, Mok U, Pec M, Evans B (2018b) Can we use nanoindentation to derive the poroelastic parameters of microporous rocks?-Experimental evidence. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Mighani S, Bernabé Y, Boulenouar A, Mok U, Evans B (2019) Creep deformation in Vaca Muerta shale from nanoindentation to triaxial experiments. J Geophys Res Solid Earth 124(8):7842–7868

Mikowski A, Soares P, Wypych F, Lepienski CM (2008) Fracture toughness, hardness, and elastic modulus of kyanite investigated by a depth-sensing indentation technique. Am Miner 93(5–6):844–852

Miller M, Bobko C, Vandamme M, Ulm F-J (2008) Surface roughness criteria for cement paste nanoindentation. Cem Concr Res 38(4):467–476

Monclús MA, Lotfian S, Molina-Aldareguía JM (2014) Tip shape effect on hot nanoindentation hardness and modulus measurements. Int J Precis Eng Manuf 15(8):1513–1519

Mound B, Pharr G (2019) Nanoindentation of fused quartz at loads near the cracking threshold. Exp Mech 59(3):369–380

Nadukuru SS (2013) Static fatigue: a key cause of time effects in sand, civil engineering. University of Michigan, Ann Arbor, p 242

Nguyen QD, Chung K-H (2019) Effect of tip shape on nanomechanical properties measurements using AFM. Ultramicroscopy 202:1–9

Nowak JD, Rzepiejewska-Malyska KA, Major RC, Warren OL, Michler J (2010) In-situ nanoindentation in the SEM. Mater Today 12:44–45

Oliver WC, Pharr GM (1992) An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments. J Mater Res 7(6):1564–1583

Oliver WC, Pharr GM (2004) Measurement of hardness and elastic modulus by instrumented indentation: advances in understanding and refinements to methodology. J Mater Res 19(1):3–20

Ortega JA, Ulm F-J, Abousleiman Y (2009) The nanogranular acoustic signature of shale. Geophysics 74(3):D65–D84

Ostadhassan M, Liu K, Li C, Khatibi S (2018) Fine scale characterization of shale reservoirs: methods and challenges. Springer, Berlin

Packard C, Wheeler J, Trenkle J, Schuh C (2016) Nanoindentation: high temperature. In: Reference module in materials science and materials engineering

Pant RR (2013) Nanoindentation characterization of clay minerals and clay-based hybrid bio-geomaterials. Department of Civil and Environmental Engineering, Louisiana State University, Baton Rouge, p 111

Pant R, Hu L, Zhang G (2013) Anisotropy of mica probed by nanoindentation. In: Laloui L, Ferrari A (eds) Multiphysical testing of soils and shales. Springer, Berlin, pp 239–245

Park HK (2018) Characterisation and performance of pindan soils modified with polymer stabilisers. Civil Engineering, Curtin University, Bentley

Pathak S, Kalidindi SR (2015) Spherical nanoindentation stress–strain curves. Mater Sci Eng R Rep 91:1–36

Pathak S, Shaffer J, Kalidindi SR (2009) Determination of an effective zero-point and extraction of indentation stress–strain curves without the continuous stiffness measurement signal. Scr Mater 60(6):439–442

Peng ZJ, Wen T, Gong JH, Wang CB, Fu ZQ, Miao HZ (2012) Relationship between the ratio of Young's modulus to hardness and the elastic recovery of nanoindentation. In: Key engineering materials. Trans Tech Publ, pp 5–8

Penumadu D, Dutta AK, Luo X, Thomas KG (2009) Nano and neutron science applications for geomechanics. KSCE J Civ Eng 13(4):233–242

Peykov D, Martin E, Chromik RR, Gauvin R, Trudeau M (2012) Evaluation of strain rate sensitivity by constant load nanoindentation. J Mater Sci 47(20):7189–7200

Phani PS, Oliver W, Pharr G (2017) On the measurement of power law creep parameters from instrumented indentation. JOM 69(11):2229–2236

Pharr G (1998) Measurement of mechanical properties by ultra-low load indentation. Mater Sci Eng, A 253(1–2):151–159

Prach O, Minnert C, Johanns KE, Durst K (2019) A new nanoindentation creep technique using constant contact pressure. J Mater Res 34(14):2492–2500

Qiu Y (2019) Upscaling micromechanical indentation properties to mesoscale behaviour in geomaterials. Department of Civil Engineering, University of Toronto, Toronto, p 96

Randall NX, Vandamme M, Ulm F-J (2009) Nanoindentation analysis as a two-dimensional tool for mapping the mechanical properties of complex surfaces. J Mater Res 24(3):679–690

Raza A, Gholami R, Rabiei M, Rasouli V, Rezaee R, Fakhari N (2019) Impact of geochemical and geomechanical changes on CO_2 sequestration potential in sandstone and limestone aquifers. Greenh Gases Sci Technol 9(5):905–923

Saenger EH, Lebedev M, Uribe D, Osorno M, Vialle S, Duda M, Iglauer S, Steeb H (2016) Analysis of high-resolution X-ray computed tomography images of Bentheim sandstone under elevated confining pressures. Geophys Prospect 64(4):848–859

Sahoo RK, Rout PP, Singh SK, Mishra BK, Mohapatra BK (2017) Synergetic surface and chemical durability study of the aesthetically enhanced natural quartz by heat treatment. Metall Mater Trans A 48(3):1111–1120

Schuh CA, Packard CE, Lund AC (2006) Nanoindentation and contact-mode imaging at high temperatures. J Mater Res 21(3):725–736

Sebastiani M, Bemporad E, Carassiti F, Schwarzer N (2011) Residual stress measurement at the micrometer scale: focused ion beam (FIB) milling and nanoindentation testing. Philos Mag 91(7–9):1121–1136

Sebastiani M, Johanns KE, Herbert EG, Pharr GM (2015a) Measurement of fracture toughness by nanoindentation methods: recent advances and future challenges. Curr Opin Solid State Mater Sci 19(6):324–333

Sebastiani M, Johanns K, Herbert EG, Carassiti F, Pharr GM (2015b) A novel pillar indentation splitting test for measuring fracture toughness of thin ceramic coatings. Philos Mag 95(16–18):1928–1944

Sharma P, Prakash R, Abedi S (2019) Effect of temperature on nano-and microscale creep properties of organic-rich shales. J Pet Sci Eng 175:375–388

Shi X, Yang L, Li D, Ding X (2018) Mechanical characterization of Longmaxi Marine Shale by nanoindentation. In: ISRM international symposium-10th Asian rock mechanics symposium, international society for rock mechanics and rock engineering

Shi X, He Z, Long S, Peng Y, Li D, Jiang S (2019) Loading rate effect on the mechanical behavior of brittle longmaxi shale in nanoindentation. Int J Hydrog Energy 44(13):6481–6490

Shi X, Jiang S, Wang Z, Bai B, Xiao D, Tang M (2020a) Application of nanoindentation technology for characterizing the mechanical properties of shale before and after supercritical CO_2 fluid treatment. J CO2 Util 37:158–172

Shi X, Jiang S, Yang L, Tang M, Xiao D (2020b) Modeling the viscoelasticity of shale by nanoindentation creep tests. Int J Rock Mech Min Sci 127:104210

Shukla P (2013) Nanoindentation studies on shales. Mewbourne school of petroleum and geological engineering, University of Oklahoma, Norman

Shukla P, Kumar V, Curtis M, Sondergeld CH, Rai CS (2013) Nanoindentation studies on shales. In: 47th U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, San Francisco, California, p 10

Shukla P, Taneja S, Sondergeld C, Rai C (2015) Nanoindentation measurements on rocks, fracture, fatigue, failure, and damage evolution, vol 5. Springer, Berlin, pp 99–105

Singh DN (2019) Novel techniques to simulate and monitor contaminant-geomaterial interactions. Indian Geotech J 49(1):2–36

Skrzypczak M, Guerret-Piecourt C, Bec S, Loubet J-L, Guerret O (2009) Use of a nanoindentation fatigue test to characterize the ductile–brittle transition. J Eur Ceram Soc 29(6):1021–1028

Slim MI (2016) Creep properties of source rocks using indentation: the role of organic matter on texture and creep rates. Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, p 404

Slim M, Abedi S, Bryndzia LT, Ulm F-J (2018) Role of organic matter on nanoscale and microscale creep properties of source rocks. J Eng Mech 145(1):04018121

Sly MK, Thind AS, Mishra R, Flores KM, Skemer P (2019) Low-temperature rheology of calcite. Geophys J Int 221:129–141. https://doi.org/10.1093/gji/ggz577

Su D, Li X (2014) Investigation of near-surface mechanical properties of materials using atomic force microscopy. Exp Mech 54(1):11–24

Su C, Herbert EG, Sohn S, LaManna JA, Oliver WC, Pharr GM (2013) Measurement of power-law creep parameters by

instrumented indentation methods. J Mech Phys Solids 61(2):517–536
Su X, Chen P, Ma T (2019) Evaluation of shale fracture toughness based on micrometer indentation test. Petroleum 5(1):52–57
Sudharshan Phani P, Oliver WC (2019) A critical assessment of the effect of indentation spacing on the measurement of hardness and modulus using instrumented indentation testing. Mater Des 164:107563
Takagi H, Fujiwara M (2014) Set of conversion coefficients for extracting uniaxial creep data from pseudo-steady indentation creep test results. Mater Sci Eng, A 602:98–104
Tang Z, Guo Y, Jia Z, Li Y, Wei Q (2015) Examining the effect of pileup on the accuracy of sharp indentation testing. Adv Mater Sci Eng 2015. https://doi.org/10.1155/2015/528729
Tankiewicz M (2018) Application of the nanoindentation technique for the characterization of varved clay. Open Geosci 10(1):902–910
Thom CA (2019) Scale-dependent plasticity and nanoindentation creep of geologic materials. University of Pennsylvania, Philadelphia
Thom CA, Goldsby DL (2019) Nanoindentation Studies of Plasticity and Dislocation Creep in Halite. Geosciences 9(2):79
Thom C, Brodsky E, Carpick R, Pharr G, Oliver W, Goldsby D (2017) Nanoscale roughness of natural fault surfaces controlled by scale-dependent yield strength. Geophys Res Lett 44(18):9299–9307
Thom C, Carpick R, Goldsby D (2018) Constraints on the physical mechanism of frictional aging from nanoindentation. Geophys Res Lett 45(24):13306–13311
Tian S, Zhang P, Sheng M, Wang T, Tang J, Xiao L (2020) Modification of microscopic properties of shale by carbonic acid treatment: implications for CO_2-based fracturing in Shale Formations. Energy Fuels 34:3458–3466. https://doi.org/10.1021/acs.energyfuels.9b03772
Toyama H, Kishida H, Yonezu A (2018) Characterization of fatigue crack growth of concrete mortar under cyclic indentation loading. Eng Fail Anal 83:156–166
Tyurin AI, Victorov SD, Kochanov AN, Shuklinov AV, Pirozhkova TS (2016) Methods of micro-and nanoindentation for characterization of local physical and mechanical properties of multiphase materials. In: AIP conference proceedings. AIP Publishing, p 020227
Ulm F-J, Abousleiman Y (2006) The nanogranular nature of shale. Acta Geotech 1(2):77–88
Valdes CC, Heidari Z (2017) Application of nanoindentation for uncertainty assessment of elastic properties in mudrocks from micro-to well-log scales. Geophysics 82(6):D327–D339
Vandamme M, Ulm F-J (2013) Nanoindentation investigation of creep properties of calcium silicate hydrates. Cem Concr Res 52:38–52
Vandamme M, Tweedie CA, Constantinides G, Ulm F-J, Van Vliet KJ (2012) Quantifying plasticity-independent creep compliance and relaxation of viscoelastoplastic materials under contact loading. J Mater Res 27(1):302–312
Veytskin YB, Tammina VK, Bobko CP, Hartley PG, Clennell MB, Dewhurst DN, Dagastine RR (2017) Micromechanical characterization of shales through nanoindentation and energy dispersive x-ray spectrometry. Geomech Energy Environ 9:21–35
Vialle S, Lebedev M (2015) Heterogeneities in the elastic properties of microporous carbonate rocks at the microscale from nanoindentation tests. SEG technical program expanded abstracts 2015. Society of Exploration Geophysicists, pp 3279–3284
Vialle S, Pazdniakou A, Adler P (2018) Prediction of the macroscopic mechanical properties of carbonate from nano-indentation tests. In: 52nd US rock mechanics/geomechanics symposium. American Rock Mechanics Association
Viktorov S, Golovin YI, Kochanov A, Tyurin A, Shuklinov A, Shuvarin I, Pirozhkova T (2014) Micro-and nano-indentation approach to strength and deformation characteristics of minerals. J Min Sci 50(4):652–659
Wang L, Liu X (2019) Correlation analysis of surface tilt effect on its mechanical properties by nano-indentation. Int J Precis Eng Manuf 20(3):327–335
Wang L, Zhang Y (2018) Elastic properties estimation of Longmaxi Shale in Sichuan Basin with nanoindentation. In: International geophysical conference, Beijing, China, 24–27 April 2018, Society of Exploration Geophysicists and Chinese Petroleum Society, pp 1232–1235
Wang F, Fu B, Mirshams RA, Cooper W, Komanduri R, Lu H (2011) Mechanical properties measurement of sand grains by nanoindentation. In: Proulx T (ed) Time dependent constitutive behavior and fracture/failure processes, vol 3. Springer, pp 121–130
Wang J, Hodgson PD, Yang C (2012) Effects of mechanical properties on the contact profile in Berkovich nanoindentation of elastoplastic materials. J Mater Res 27(1):313–319
Wang F, Fu B, Luo H, Staggs S, Mirshams R, Cooper W, Park S, Kim M, Hartley C, Lu H (2014) Characterization of the grain-level mechanical behavior of Eglin sand by nanoindentation. Exp Mech 54(5):871–884
Wang S, Liu K, Han J, Ling K, Wang H, Jia B (2019a) Investigation of properties alternation during super-critical CO_2 injection in shale. Appl Sci 9(8):1686
Wang S, Xu H, Wang Y, Kong L, Wang Z, Liu S, Zhang J, Zhao H (2019b) Design and testing of a cryogenic indentation apparatus. Rev Sci Instrum 90(1):015117
Wheeler J, Michler J (2013) Invited article: indenter materials for high temperature nanoindentation. Rev Sci Instrum 84(10):101301
Wheeler J, Armstrong D, Heinz W, Schwaiger R (2015) High temperature nanoindentation: the state of the art and future challenges. Curr Opin Solid State Mater Sci 19(6):354–366
Whitney DL, Broz M, Cook RF (2007) Hardness, toughness, and modulus of some common metamorphic minerals. Am Miner 92(2–3):281–288
Wilkinson TM (2014) High-resolution, mechanical property mapping in oil shales. Metallurgical and Materials Engineering, Colorado School of Mines, p 75
Wilkinson TM, Zargari S, Prasad M, Packard CE (2015) Optimizing nano-dynamic mechanical analysis for high-resolution, elastic modulus mapping in organic-rich shales. J Mater Sci 50(3):1041–1049
Xian S, Jiang S, Shuangfang L, Zhiliang H, Dongjie L, Zhixuan W, Dianshi X (2019) Investigation of mechanical

properties of bedded shale by nanoindentation tests: a case study on Lower Silurian Longmaxi Formation of Youyang area in southeast Chongqing. China. Pet Explor Dev 46(1):163–172

Xiang D, Chen Z, Yang Z, Wang S, Wang Q, Hou D, Zhang G (2017) Probing the mechanical properties of shales by nanoindentation. In: Brandon TL, Valentine RJ (eds) Geotechnical frontiers. pp 497–507

Xiao J, Ding D, Xu G, Jiang F (2008) Waveform effect on quasi-dynamic loading condition and the mechanical properties of brittle materials. Int J Rock Mech Min Sci 45(4):621–626

Xie Q (2016) Mechanical characterization of kerogen in black siliceous shale via nanoindentation. SEG technical program expanded abstracts 2016, Society of Exploration Geophysicists, pp 3348–3353

Xu Z-H, Li X (2007) Effect of sample tilt on nanoindentation behaviour of materials. Philos Mag 87(16):2299–2312

Xu Z-H, Li X (2008) Residual stress determination using nanoindentation technique. In: Yang F, Li JCM (eds) Micro and nano mechanical testing of materials and devices. Springer, Berlin, pp 136–150

Xu T, Du Y, Luo H, Hu Z, Wang X, Guo L, Lu H (2018) Characterization of the mechanical behavior of colorado mason sand at grain-level by nanoindentation. Exp Mech 58(3):449–463

Yang Z, Wang L, Zhang G, Ho C (2016) Micromechanical characterization of fluid–shale interactions via nanoindentation. In: SPE Asia pacific hydraulic fracturing conference. Society of Petroleum Engineers

Yang H, Luo S, Zhang G, Song J (2017) Elasticity of clay shale characterized by nanoindentation. In: 51st U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, San Francisco, California, USA, p 7

Yang Z, Wang L, Chen Z, Xiang D, Hou D, Ho CL, Zhang G (2018) Micromechanical characterization of fluid/shale interactions by means of nanoindentation. SPE Reservoir Eval Eng 21(02):405–417

Yang S-Q, Yin P-F, Huang Y-H (2019) Experiment and discrete element modelling on strength, deformation and failure behaviour of shale under Brazilian compression. Rock Mech Rock Eng 52(11):4339–4359

Yang S-Q, Yin P-F, Ranjith P (2020) Experimental Study on mechanical behavior and brittleness characteristics of Longmaxi Formation Shale in changning, China. Rock Mechanics and Rock Engineering, Sichuan Basin, pp 1–23

Yin H, Zhang G (2011a) Nanoindentation behavior of muscovite subjected to repeated loading. J Nanomech Micromech 1(2):72–83

Yin H, Zhang G (2011b) Cyclic nanoindentation shakedown of muscovite and its elastic modulus measurement, MEMS and nanotechnology, vol 4. Springer, Berlin, pp 83–92

Yu H, Zhang Y, Lebedev M, Han T, Verrall M, Wang Z, Al-Khdheeawi E, Al-Yaseri A, Iglauer S (2018) Nanoscale geomechanical properties of Western Australian coal. J Pet Sci Eng 162:736–746

Zargari S (2015) Effect of thermal maturity on nanomechanical properties and porosity in organic rich shales (a Bakken shale case study). Petroleum Engineering, Colorado School of Mines, Golden, p 135

Zargari S, Prasad M, Mba KC, Mattson E (2011) Organic maturity, hydrous pyrolysis, and elastic property in shales. In: Canadian unconventional resources conference. Society of Petroleum Engineers

Zargari S, Prasad M, Mba KC, Mattson ED (2013) Organic maturity, elastic properties, and textural characteristics of self resourcing reservoirs. Geophysics 78(4):D223–D235

Zargari S, Wilkinson TM, Packard CE, Prasad M (2016) Effect of thermal maturity on elastic properties of kerogen. Geophysics 81(2):M1–M6

Zeng Q, Feng Y, Xu S (2017) A discussion of "Application of nano-indentation methods to estimate nanoscale mechanical properties of shale reservoir rocks" by K Liu, M Osatadhassan and B Bubach. J Nat Gas Sci Eng 42:187–189

Zeszotarski JC, Chromik RR, Vinci RP, Messmer MC, Michels R, Larsen JW (2004) Imaging and mechanical property measurements of kerogen via nanoindentation. Geochim Cosmochim Acta 68(20):4113–4119

Zhang G, Wei Z, Ferrell RE (2009) Elastic modulus and hardness of muscovite and rectorite determined by nanoindentation. Appl Clay Sci 43(2):271–281

Zhang G, Wei Z, Ferrell RE, Guggenheim S, Cygan RT, Luo J (2010) Evaluation of the elasticity normal to the basal plane of non-expandable 2: 1 phyllosilicate minerals by nanoindentation. Am Miner 95(5–6):863–869

Zhang J, Hu L, Pant R, Yu Y, Wei Z, Zhang G (2013) Effects of interlayer interactions on the nanoindentation behavior and hardness of 2: 1 phyllosilicates. Appl Clay Sci 80:267–280

Zhang Y, Lebedev M, Sarmadivaleh M, Barifcani A, Iglauer S (2016) Change in geomechanical properties of limestone due to supercritical CO_2 injection. In: SPE Asia pacific oil and gas conference and exhibition. Society of Petroleum Engineers, Perth, Australia, p 10

Zhang Y, Zhang Z, Sarmadivaleh M, Lebedev M, Barifcani A, Iglauer S (2017) Prediction of microscale rock mechanical performance from microCT images: heterogeneous coal as an example. In: 51st US rock mechanics/geomechanics symposium. American Rock Mechanics Association

Zhang F, Guo H, Hu D, Shao J-F (2018a) Characterization of the mechanical properties of a claystone by nano-indentation and homogenization. Acta Geotech 13(6):1395–1404

Zhang Y, Lebedev M, Al-Yaseri A, Yu H, Nwidee LN, Sarmadivaleh M, Barifcani A, Iglauer S (2018b) Morphological evaluation of heterogeneous oolitic limestone under pressure and fluid flow using X-ray microtomography. J Appl Geophys 150:172–181

Zhang F, Hu W, Guo HQ, Hu DW, Sheng Q, Shao JF (2018c) Nanoindentation tests on granite after heat treatment. Rock Soil Mech 39:235–243 (In Chinese). https://doi.org/10.16285/j.rsm.2018.0052

Zhang Y, Lebedev M, Iglauer S (2018d) The nanoscale rock mechanical properties of different rank coal by nanoindentation test. In: 52nd U.S. rock mechanics/geomechanics symposium. American Rock Mechanics Association, Seattle, Washington, p 5

Zhang Y, Lebedev M, Al-Yaseri A, Yu H, Xu X, Iglauer S (2018e) Characterization of nanoscale rockmechanical properties and microstructures of a Chinese sub-bituminous coal. J Nat Gas Sci Eng 52:106–116

Zhang Y, Lebedev M, Al-Yaseri A, Yu H, Xu X, Sarmadivaleh M, Barifcani A, Iglauer S (2018f) Nanoscale rock mechanical property changes in heterogeneous coal after water adsorption. Fuel 218:23–32

Zhang P-P, Tian S-C, Sheng M, Wang T-Y, Khan WA, Xu Q, Xiao L-Z (2019) Effect of dilute acid treatment on adhesion properties of Longmaxi black shale. Pet Sci 16(6):1320–1331

Zhao J, Zhang D, Wu T, Tang H, Xuan Q, Jiang Z, Dai C (2018) Multiscale approach for mechanical characterization of organic-rich shale and its application. Int J Geomech 19(1):04018180

Zhu W, Hughes JJ, Bicanic N, Pearce CJ (2007) Nanoindentation mapping of mechanical properties of cement paste and natural rocks. Mater Charact 58(11–12):1189–1198

Zhu W, Fonteyn MTJ, Hughes J, Pearce C (2009) Nanoindentation study of resin impregnated sandstone and early-age cement paste specimens. In: Bittnar Z, Bartos PJM, Němeček J, Šmilauer V, Zeman J (eds) Nanotechnology in construction 3. Springer, Berlin Heidelberg, Berlin, Heidelberg, pp 403–408

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